

A Systematic Literature Review of Adaptive Parameter Control Methods for Evolutionary Algorithms

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Evolutionary algorithms (EAs) are robust stochastic optimisers that perform well over a wide range of problems. Their robustness, however, may be affected by several adjustable parameters, such as mutation rate, crossover rate, and population size. Algorithm parameters are usually problem-specific, and often have to be tuned not only to the problem but even the problem instance at hand to achieve ideal performance. In addition, research has shown that different parameter values may be optimal at different stages of the optimisation process. To address these issues, researchers have shifted their focus to adaptive parameter control, in which parameter values are adjusted during the optimisation process based on the performance of the algorithm. These methods redefine parameter values repeatedly based on implicit or explicit rules that decide how to make the best use of feedback from the optimisation algorithm.

In this survey, we systematically investigate the state of the art in adaptive parameter control. The approaches are classified using a new conceptual model that subdivides the process of adapting parameter values into four steps that are present explicitly or implicitly in all existing approaches that tune parameters dynamically during the optimisation process. The analysis reveals the major focus areas of adaptive parameter control research as well as gaps and potential directions for further development in this area.

CCS Concepts: • **Computing methodologies** → *Search methodologies*;

Additional Key Words and Phrases: Evolutionary algorithms, adaptive parameter control

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1. INTRODUCTION

Evolutionary algorithms (EAs) have been used to solve problems in cases for which exact approaches could not be devised, and to find approximate solutions when optimal results are hard to achieve [Goldberg 1989]. The success of EA applications, however, depends on the parameter settings, such as the choice of representation, parent selection, recombination and mutation operators [Eiben and Smit 2012], which have to be configured for every new problem instance. It has been argued that setting parameters does not guarantee optimal algorithm behaviour, since different algorithm configurations may be optimal at different stages of the optimisation process [Goldman and Tauritz 2011; Bäck 1992; Stephens et al. 1998; Smith and Fogarty 1996; Hesser and Manner 1991]. An intuitive explanation is the dynamic behaviour of the optimisation

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process, which evolves from a more diffused global search, requiring parameter values suited for the exploration of the search space, to a more focused local optimisation process, which requires parameter values that help with the convergence of the algorithm. For instance, Cervantes and Stephens [2009] showed that any fixed mutation rate can be expected to be suboptimal in terms of balancing exploration and exploitation. Parameter control also has the advantage of adapting the search strategy in dynamic landscapes, where the fitness landscape may change over time, which requires new parameter settings for optimal algorithm performance. Perhaps the most practical benefit of parameter control is the fact that it removes the necessity to conduct time-consuming and laborious explorations of the large space of possible parameter value combinations whenever the algorithm is applied to new problem instances.

The distinguishing feature of different parameter control methods is the way that parameters are adapted during the search process. Rules for changing parameter values can be classified into three categories: uninformed, self-adaptive, and adaptive parameter control [Eiben et al. 2007a]. Uninformed parameter control [Eiben and Smith 2004] chooses the ideal parameter settings based on time, without using feedback from the search [Mezura-Montes and Palomeque-Ortiz 2009; Bäck and Schütz 1996], similar to the way that the cooling schedule is applied in simulating annealing. The problem then lies in devising such a schedule based on the total number of iterations that the optimisation algorithm is likely to take before it converges, which is hard to predict before the run.

Self-adaptive parameter control combines the search for optimal parameter values with the search for optimal solutions. This is usually implemented by encoding parameter values into the solution genotype, which are evolved simultaneously [Bäck et al. 2000; Farmani and Wright 2003; Deb and Beyer 2001]. High-fitness solutions survive to the next generation and propagate their superior traits along with the parameter values assumed responsible for high quality. Including the parameter space in the search space extends the solution size and increases the complexity of the optimisation problem. Furthermore, the search for optimal parameter values may be subject to premature convergence, which is a common problem faced by the search for optimal solutions [Cervantes and Stephens 2009; Meyer-Nieberg and Beyer 2007].

Adaptive parameter control separates the search for optimal solutions from the search for ideal parameter values [Thierens 2005; Fialho et al. 2009; Corne et al. 2002; Davis 1989; Hong et al. 2000; Igel and Kreutz 2003; Julstrom 1995; Lobo and Goldberg 1997; Dirk Schlierkamp-Voosen 1994; Tuson and Ross 1998]. These methods monitor properties of an EA run, such as the quality of the solutions. The change in algorithm properties is used as a guide to adjust parameter values in the following iterations. Unlike in self-adaptive parameter control, the update mechanism and the selection of the ideal parameter values are not part of the optimisation cycle.

The field of parameter control has been widely researched in recent years [Hansen and Ostermeier 2001; Fialho et al. 2009; Thierens 2005]. Comprehensive reviews that investigate research trends in parameter control methods in general indicate an increasing interest in the area [Karafotias et al. 2014b; Eiben et al. 1999; Eiben and Schut 2008; Smith and Fogarty 1997; Eiben et al. 2007; DeJong 2007; Angeline 1995]. More specialised literature reviews have focused on specific algorithm parameters, such as methods that adjust penalty techniques [Barbosa et al. 2015] and population sizing approaches [Eiben et al. 2004; Lobo and Lima 2007, 2005]. Research activities on similar topics are often carried out independently by different research communities. In this study, we aim to connect the knowledge and build a gateway to new approaches for adaptive parameter control. To this end, we perform a systematic literature review (SLR) on 152 papers in the area of adaptive parameter control. In contrast to an expert review using ad-hoc literature selection, an SLR is a methodologically rigorous review

of research results. The aim is to systematically curate and assess published research in the area of adaptive parameter control, and develop evidence-based guidelines for practitioners that aim to use these methods as well as researchers who wish to push the boundary of knowledge in this area. One of the main outcomes of this SLR is a new conceptual model that provides a natural division of adaptive parameter control techniques, facilitates their analysis and comparison, and helps in developing and judging new approaches. The model is then used to identify research gaps and guide future research.

2. RESEARCH METHOD

This study aims at answering the following research questions:

- RQ1** How can the current research in adaptive parameter control be classified?
- RQ2** What is the current state of adaptive parameter control research with respect to this classification?
- RQ3** What can be learnt from the current research results that will lead to topics for further investigation?

The method used to answer these questions is an SLR undertaken following the guidelines presented in CRD Report [2009]. The SLR involved three main steps: planning, conducting the review, and reporting the results.

2.1. Planning the Review

The first stage in the planning of the review involves the identification of the appropriate search engines and keyword list for the search. The search strategy for the review was primarily directed toward finding published papers in journals and conference proceedings via widely accepted search engines and databases, such as Google Scholar, IEEE Explore, ACM Digital Library, Springer Digital Library, and Elsevier ScienceDirect.

For the search, we focused on selected key search terms, including ‘adaptive parameter control’, ‘controlling parameters of evolutionary algorithms’, ‘methods for controlling EA parameters’ and ‘parameter adaptation’. Supporting search terms include ‘machine learning in evolutionary algorithms’, ‘adjusting mutation rate’, ‘adapting crossover rate’, ‘dynamic population size’, and so on. The complete list of the search terms can be found at <http://users.monash.edu.au/~aldeidaa/adaptiveParameterControl.html>. The keywords were refined and extended during the search process. Forward and backward references were examined and relevant papers were included for analysis. Other literature reviews in parameter control [Eiben et al. 1999; Eiben and Schut 2008; Smith and Fogarty 1997; Eiben et al. 2007; DeJong 2007], adaptive penalty techniques [Barbosa et al. 2015], and population size adjustment [Eiben et al. 2004; Lobo and Lima 2007, 2005] were used to cross-check for missing papers.

The keyword list may return papers that are not relevant to this literature review; thus, it is important to define a clear filtering process based on a set of inclusion and exclusion criteria. Self-adaptive and deterministic methods are not part of this literature review. Papers that introduce an adaptive method, but do not validate the approach (e.g., Cárdenas-Barrón [2010] and Harvey [1992]), were also excluded from this systematic review.

There is a large body of work in algorithm selection [Leyton-Brown et al. 2003], and hyper-heuristics [Burke et al. 2013] that do not fit in this SLR, since they consider the choice between different algorithms instead of the choice between different parameter values of an algorithm. The methods included in this work follow the strict rule that the control mechanism is used to adapt parameters of an EA, and the adaptation happens during the optimisation process. Methods that separate the learning phase from the

optimisation phase are not included, since they do not fall into the category of adaptive parameter control. In addition, the review does not include adaptive methods that adjust parameters that are not part of the main EA strategy, such as the adaptation of the subsidiary local search strategy [Wu et al. 2014; Lobo and Goldberg 1997]. Regarding the time frame, the papers included in this review range from one of the earliest methods in adaptive parameter control published in 1986 [Grefenstette 1986] to works published in 2015.

2.2. Conducting the Review

To answer the first research question (RQ1), we conducted a keyword-based collection of published papers in adaptive parameter control. First, we reviewed and filtered the papers according to the inclusion and exclusion criteria identified in the planning phase (Section 2.1). This process resulted in the inclusion of 152 papers, which were filtered from 220 initially collected papers. Next, we performed a content analysis of the papers, and derived a taxonomy. It is important that the taxonomy scheme captures the depth of knowledge recorded about each approach studied. Hence, we used an iterative coding process to identify the main categories of the taxonomy. The coding process was based on the grounded theory method [Glaser and Strauss 2009]. We analysed each paper to identify new concepts, and consolidated the taxonomy terms by merging the synonyms and unifying the terms. The output of this process is a conceptual model of adaptive parameter control approaches presented in Section 4.

2.3. Data Analysis

We reviewed each paper and classified the related work based on the taxonomy derived in the first research task. The classification of the papers was cross-checked and analysed for any inconsistencies. Extracted data was stored in a database, which enabled a descriptive quantitative analysis. The aim of the data extraction and resulting classification was to provide a significant overview of the current research efforts in this domain, answering RQ2. Section 5 presents the findings. Finally, we cross-analysed the survey results and synthesized possible directions for further research, answering the third research question (RQ3). The findings are presented in Section 6.

2.4. Threats to Validity

One of the main threats to the validity of this SLR is having an incomplete list of papers in the area of adaptive parameter control. The main contributors to this threat are the selected list of keywords and the bias from the search engines. To decrease the risk of an incomplete keyword list, we have used an iterative approach to keyword-list construction. A well-known set of papers was used to build the initial taxonomy, which evolved in iterations. New keywords were added when the keyword list would not produce new search results. In order to deal with the limitations of any particular search engine, we employed multiple search engines.

Another important issue is related to the robustness of the taxonomy derived. To avoid a taxonomy with insufficient capability to classify the selected papers, we used an iterative content analysis method to continuously evolve the taxonomy for every new concept encountered in the papers.

3. PARAMETERS CONTROLLED

Common implementations of EAs need values for *crossover and mutation rates* as well as *population size*. More sophisticated implementations may switch between a number of *crossover and mutation operators*, which induces variation in the search paths through the fitness landscape. Another crucial factor that influences the search landscapes and dictates the applicability of operators is the *representation* (encoding)

Table I. Total Number of Papers Classified Based on the Parameter Being Adapted

Parameter	Count	Percentage	Parameter	Count	Percentage
Representation	3	1.887%	Problem specific	28	17.610%
Crossover operator	21	13.208%	Crossover rate	36	22.642%
Mutation operator	29	18.239%	Mutation rate	44	27.673%
Population size	37	23.270%	Parent selection	15	9.434%
Replacement	2	1.258%	Fitness function	34	21.384%
Offspring size	3	1.887%			

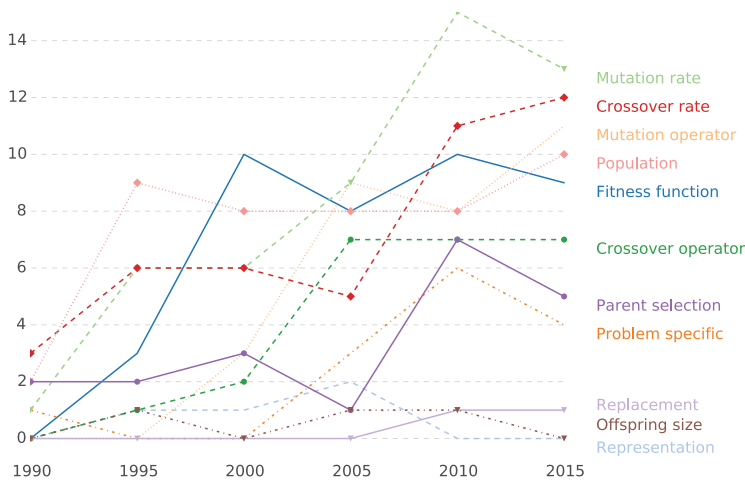


Fig. 1. Number of papers per year that discuss controlling each EA parameter.

of the solution. Variations of the *parent selection procedure* include greedy strategies, fitness-proportional strategies [Holland 1975], and tournament selection [McGinley et al. 2011]. Adaptive methods most often use the Boltzmann selection strategy [Bäck et al. 1997] to vary selection pressure. The *replacement* procedure is responsible for accepting or rejecting new solutions into the population, and its strategy affects the diversity of the population [Arabas et al. 1994]. The *fitness function* values as inputs to the algorithm have also been varied in environments in which constraints play a role and penalty functions are applied to navigate a constrained search space. EA implementations that are specific to a certain problem often introduce new parameters. A number of authors have published studies in which these custom parameters are adapted during the progress of the search.

Table I illustrates that crossover and mutation rates, as well as population size, are the most popular parameters controlled during the search, followed closely, and perhaps surprisingly, by the fitness function. Controlling parent selection is somewhat rare, and the adaptation of representation, offspring, and replacement has only been attempted a few times, no doubt because of the difficulty of implementation. An alternative approach to automating the choice of representation is meta-evolution, as introduced by Scott and Bassett [2015].

Figure 1 shows the development of the focus of research by algorithm parameter. The number of papers that adapt the fitness function spiked during 1995–2000 and has remained steady since. The interest in controlling mutation and crossover rates is still growing, and most studies discuss the adaptation of both parameters.

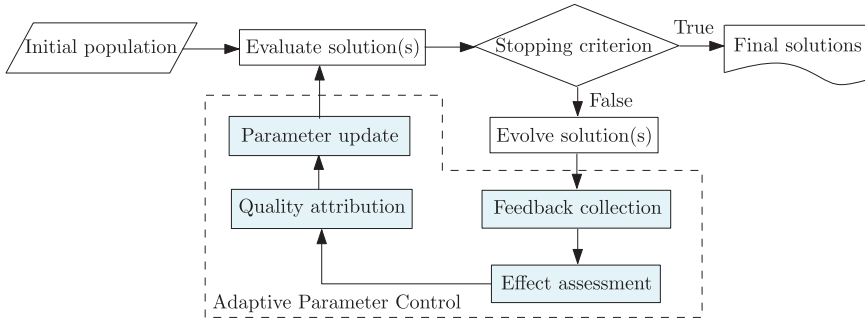


Fig. 2. A conceptual model of adaptive parameter control in EAs.

4. A CONCEPTUAL MODEL OF ADAPTIVE PARAMETER CONTROL METHODS

Parameter control addresses the requirement of finding optimal parameter configurations as the search proceeds. It describes a process in which optimisation starts with suboptimal parameter values that are adapted during the progress of the algorithm. The set of algorithm parameters that have to be adjusted is formally defined as $\{v_1, \dots, v_n\}$, where n is the number of parameters. Each parameter v_i has $\{v_{i1}, \dots, v_{im}\}$ values, where m may be the number of discrete values or the upper bound of a continuous interval. The parameter control derives the best next parameter value v_{ij} such that the influence of v_i on the performance of the algorithm is optimised.

For instance, in the case in which v_1 represents the mutation operator, v_{11} may stand for the single-point mutation operator, whereas v_{12} may represent the uniform mutation operator. In the continuous case of optimising the mutation rate v_2 , v_{22} symbolises the second range of the interval from which the mutation rate is sampled.

The conceptual model for adaptive parameter control derived using an iterative coding process is shown in Figure 2. The shaded boxes represent the four main steps that are performed when adapting parameter values of an EA: (i) feedback collection, (ii) effect assessment, (iii) quality attribution, and (iv) parameter update. The steps are interdependent, which means that the success of each step depends on the other steps.

4.1. Feedback Collection

The feedback collection strategy is a distinguishing feature of adaptive parameter control, since other parameter control methods either do not consider any feedback from the search (uninformed parameter control) or the feedback is implicit (self-adaptive). The feedback is a measure of certain properties of an EA, which are an indication of the algorithm's behaviour. The interpretation of the quality of the parameter values depends on the information provided by the feedback collection strategy, which makes it a very important step of adaptive parameter control. The analysis of the literature resulted in five different feedback categories:

- i) *Phenotype feedback* refers to the observed behaviour or quality of a solution. The majority of the approaches use the fitness expression of the phenotype. In some problems, it is hard or even impossible to define the fitness function. In these cases, a simulation may be used to determine the fitness function value of a phenotype.
- ii) *Relative phenotype feedback* refers to the relative observed behaviour or quality of solutions, usually measured as entropy.
- iii) *Genotype feedback* refers to the information regarding the solution components of a single solution.

- iv) *Genotype diversity feedback* is the diversity of the solution components in a population of solutions.
- v) *Feasibility feedback* is a measure of the violations of constraints.

4.2. Effect Assessment

The parameter effect assessment strategy derives the effect of parameter values based on the output from the feedback collection strategy. At each time step, the effect of each parameter value on the performance of the algorithm is estimated using the parameter effect assessment strategy. The vector of all parameter effects is denoted as $\vec{e} = \{e(v_{11}), e(v_{12}), \dots, e(v_{1m_1}), \dots, e(v_{nm_n})\}$. The main goal of adaptive parameter control strategies is to adapt the vector of probabilities $\vec{p}$ such that the expected value of the cumulative effect $E[\vec{e}] = \sum_{j=1}^n e(v_{ij})$ is maximised.

The main difference between the available parameter effect assessment methods is how they define the ‘success’ of parameter values. Most of these methods calculate the effect of parameter values using the quality difference of the generated solutions compared to their parents [Igel et al. 2005; Hong et al. 2002], the overall best solution [Davis 1989; Giger et al. 2007], a specific quantile (e.g., the median) [Julstrom 1995, 1997], or outlier solutions [Whitacre et al. 2006]. The iterative coding procedure, which included all papers in this literature review, extracted the following six subcategories of parameter effect assessment:

- i) *Ancestor effect*: Measured as the difference in properties (e.g., fitness) of the current solution(s) and its (their) parent(s).
- ii) *Population effect*: Measured as the change in the properties of the current solution(s) with respect to the population.
- iii) *Best effect*: Calculated as the change in the properties of the current solution(s) with respect to the best solution in the population.
- iv) *Worst effect*: The worst solution in the population is used as a reference to measure the change in the properties of the current solution(s).
- v) *Median effect*: The median solution in the population is used as a reference to measure the change in the properties of the current solution(s).
- vi) *Current effect*: The properties of the current solution(s) are used directly as the effect of the parameter value (e.g., the fitness of the solution(s)).

The term ‘current solution(s)’ refers to the solution(s) that were produced by applying a certain parameter value immediately before its effect is measured.

4.3. Quality Attribution

To make a better judgement on what would be a successful parameter value for the next iteration(s), current adaptive parameter control methods define a quality measure, which is calculated based on predefined rules that use the effect measured in the previous iterations. Whenever a parameter value is used, its current quality estimate $q(v_{i,j})$ is updated by the parameter quality attribution strategy. The vector of quality estimates for all parameter values is denoted as $\vec{q} = \{q(v_{11}), q(v_{12}), \dots, q(v_{1m_1}), \dots, q(v_{nm_n})\}$.

Some of the methods reviewed assume that the change in the solution properties is directly related to the use of certain parameter values. These methods are classified as immediate quality attribution strategies. For instance, if a mutation rate of 0.01 produces a solution that changes the fitness of its parent by Δf , the fitness improvement Δf is considered as the quality of this mutation rate. If more than one solution is created, the sum or average of the change in the properties of all solutions is attributed as the overall quality.

Ideally, quality attribution methods should consider the stochastic behaviour of EAs, which introduces a level of uncertainty in the effect of parameter values. Other methods *learn* the quality of a parameter value as the average effect [Hong et al. 2002; Igel et al. 2005], or as the best value in the last W iterations [Fialho et al. 2010], with the goal of reducing noise and uncertainty. In essence, the taxonomy derived has four different quality attribution categories. ‘Current solution(s)’ refers to the solution(s) produced by applying a certain parameter value whose effect is being measured.

- i) *Immediate quality attribution* refers to the properties of the current solution.
- ii) *Average quality attribution* considers the average change in the properties of all (or a particular subset of) solutions in the population.
- iii) *Extreme quality attribution* refers to outliers, or the best change in the properties of the current solution(s). Many methods use statistical tests to derive these outliers.
- iv) *Learned quality attribution*: The quality of parameter values is inferred using either forecasting [Aleti et al. 2014], machine-learning [Aleti and Grunske 2015], or artificial intelligence techniques [Aleti 2014].

4.4. Parameter Update

The final stage of adaptive parameter control is the update of parameter values to use in the next iteration. The update is based on the trade-off between using parameter values with high quality and exploring new values. The update mechanism used in adaptive parameter control methods can be classified into the following four categories:

- i) *Quality proportionate update*: A probability vector is kept for each parameter value, which is updated based on estimated quality, and determines how frequently a parameter value is used in future iterations.
- ii) *Quality proportionate with minimum probability update*: These methods are similar to the quality proportionate update mechanism, with the difference being that a minimum probability is assigned to each parameter value. This is to ensure that parameter values that do not perform well are not lost, since they may prove successful in future iterations.
- iii) *Greedy update*: Greedy mechanisms select the best parameter value to use in the next iteration.
- iv) *Deterministic update*: Methods that are part of this category use a deterministic rule for updating parameter values to use in the next iteration. Quality of parameter values is a direct indicator of what the next value should be.

4.5. Example

To illustrate the framework, consider the Integrated-Adaptive Genetic Algorithm (IAGA) [Luchian and Gheorghies 2003], which adapts genetic operators and their rates. The feedback collection strategy considers the fitness of the solutions; thus, it falls into the phenotype feedback category. Two different effect assessment strategies are considered: the effect on the fitness of the solution with respect to the (i) best solution from the previous generation (best-solution effect), and (ii) its parents (ancestor-solution effect). A reinforcement-based rate adaptation model is used to estimate the quality of the parameter values; thus, the approach is classified as a learned quality. The operator rates to be used for the next generation are set proportional with the reinforcement values, which qualifies the approach for the quality proportionate update category.

5. STATE OF THE ART IN ADAPTIVE PARAMETER CONTROL

The different stages of the approaches evaluated in this article are analysed based on the conceptual framework introduced in Section 4. A summary of the results is

Table II. All Methods Classified Based on Control Strategy

Feedback collection			Effect assessment		
Phenotype	137	86.164%	Ancestor	41	25.786%
Relative phenotype	9	5.660%	Population	37	23.270%
Genotype	24	15.094%	Best	55	34.591%
Genotype diversity	10	6.289%	Worst	15	9.434%
Feasibility	25	15.723%	Median	2	1.258%
			Current	47	29.560%
Quality attribution			Parameter update		
Immediate	55	34.591%	Quality-proportionate	54	33.962%
Average	21	13.208%	Quality-proportionate with min prob	21	13.208%
Extreme	17	10.692%	Greedy	60	37.736%
Learned	67	42.138%	Deterministic	32	20.126%

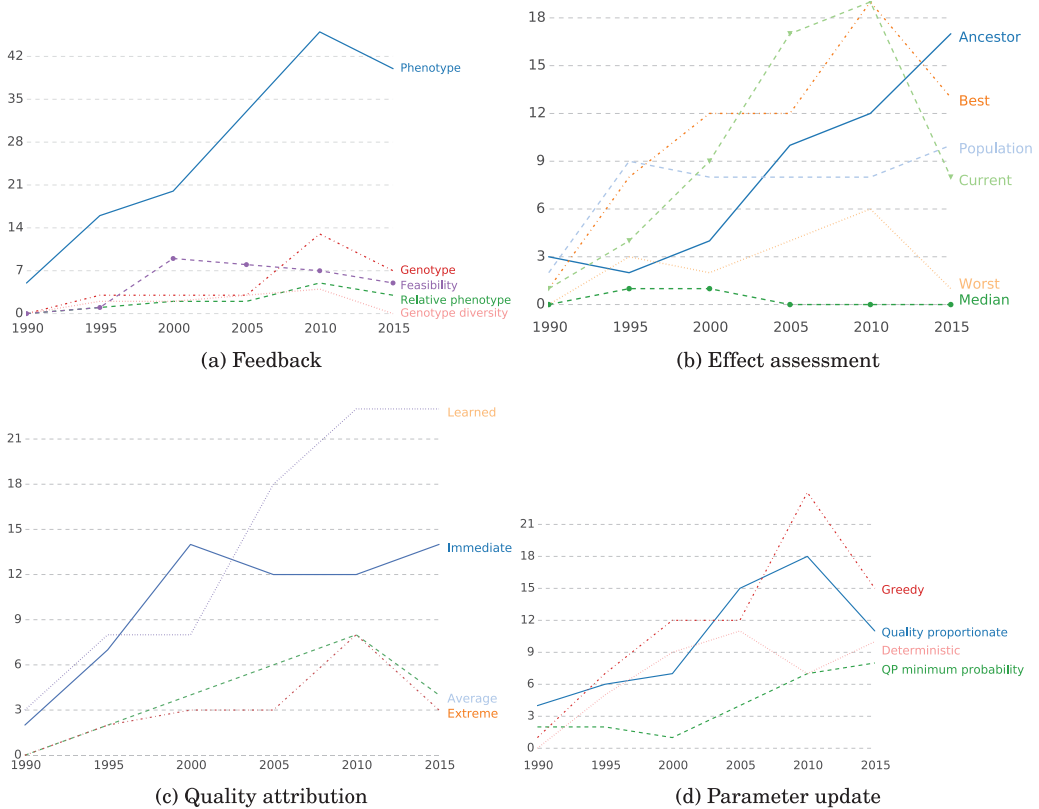


Fig. 3. Papers classified based on the adaptive parameter control mechanisms.

presented in Table II as percentage and total number of papers in each parameter control category, and Figure 3 as the development of papers over time.

The following sections discuss some of the most prominent methods from each category in detail. The aim is to answer research question RQ2: ‘What is the current state of adaptive parameter control research with respect to the taxonomy introduced in Section 4?’

5.1. Feedback Strategy

The majority of adaptive parameter control methods (more than 86% of the papers) use the fitness improvement of the solutions as an indication of the performance of the algorithm, followed by the degree of satisfaction of constraints, that is, feasibility (around 16%), and genotype information (more than 15%).

The distribution of papers in each category over the years is presented in Figure 3(a). An upward trend can be seen in the number of papers that use information regarding the phenotype as a feedback mechanism. One of the earliest research efforts in this area is an adaptive technique that rewards crossover rates that create solutions with a better fitness [Davis 1991, 1989]. The performance of the algorithm is estimated as the fitness difference between a created solution and its parent(s).

Compass [Maturana and Saubion 2008a] employs the weighted sum of the fitness improvement and diversity of the offspring, measured as the Hamming distance (i.e., genotype diversity feedback). The method calculates a vector $\vec{o}_t(s_{vij}) = (\Delta d_t(s_{vij}), \Delta f_t(s_{vij}))$ for every parameter, where $\Delta d_t(s_{vij})$ is the diversity change in solution s due to the application of parameter value v_{ij} at iteration t , and $\Delta f_t(s_{vij})$ is the fitness change.

Compass selects a vector $\vec{c}$ that has an angle θ defined in the interval $[0, \frac{\pi}{2}]$, which represents the desired balance between fitness and diversity. If θ has a value closer to zero, more importance is given to the diversity of the solutions, thus favouring exploration, whereas if θ has a value closer to $\frac{\pi}{2}$, the exploitation of good solutions becomes more important. Compass was further improved by Maturana et al. [2010] to ensure a trade-off between the fitness and diversity of the solutions, which makes θ redundant. Compass is successfully used as a feedback mechanism in ExCoDyMAB [Maturana et al. 2012], an EA with an adaptive parameter control strategy that controls variation operators. The algorithm performs exceptionally well on rough fitness landscapes, preventing the EA from converging to local optima.

Giger et al. [2007] introduced a different diversity measure that estimates the distribution of the solutions in the search space, by calculating the Euclidean distances of the solutions s_{vij} to the best solution so far s^* . First, the sum of the gene values is calculated as the Euclidean distance $d_t(s_{vij}) = \sqrt{\sum_{r=1}^g s_{vij}(r)}$, where t is the current timestep (iteration), g is the number of genes of the genotype and $s_{vij}(r)$ represents the r^{th} gene in solution s_{vij} . The authors suggest the use of a weighting factor w for each gene calculated as $d_t^w(s_{vij}) = \sqrt{\sum_{r=1}^g w_r s_{vij}(r)}$. The diversity measure for parameter value v_{ij} at iteration t is calculated as:

$$D_t(s_{vij}) = \begin{cases} 0 & \text{if } f(s_{vij}) \leq f(s^*) \\ \frac{d_t^*}{d_t^* + d_t(s_{vij})} & \text{otherwise,} \end{cases} \quad (1)$$

where d_t^* is the maximum Euclidean distance between the best solution and all solutions. In essence, a balance between the best fitness frequency and the diversity of the solutions is sought. More recently, McGinley et al. [2011] presented ACROMUSE, a novel Genetic Algorithm with adaptive crossover, mutation, and parent selection operators. The main objective of ACROMUSE is to maintain the diversity of the search space, measured as the Euclidean distance between individual solutions. The authors emphasise the importance of maintaining population diversity for increasing search coverage and for dealing with fitness landscape change.

Instead of genotype diversity, Budin et al. [1998] consider the diversity in the fitness of solutions. The degree of population diversity is estimated using the fitness of the best solutions $f(s^*)$, the worst fitness f_{min} , and the average fitness of the population $\bar{f}$

calculated as $D_t = \frac{f(s^*) - \bar{f}}{f(s^*) - f_{min}}$. Low D_t values indicate a population with similar fitness values.

Efficient exploration of the search space is one of the main motivations for using the diversity of the populations as a feedback mechanism, which is designed as an indicator of the extent to which the search space has been explored. It is important for the optimisation strategy to maintain a certain degree of diversity in the search space so that it does not get stuck in local optima. This is usually referred to as the ‘exploration–exploitation’ dilemma.

Liu et al. [2013] addressed this issue by introducing metrics that estimate the extent to which the search space has been explored and exploited, in which the search space is represented as trees. Initially, an *exploration ratio* is defined as the percentage of explored nodes over all nodes in a tree. The exploration ratio is employed to calculate the average number of generations required to change the neighbourhood, where a new neighbourhood is defined as a new and unexplored part of the search. To measure the exploitation of the tree, the average depth is calculated. The exploitation and exploration metrics are used to guide the search.

A main challenge in this area is feedback strategies for multiobjective optimisation. Generally, approaches that consider multiobjective problems focus on solution fitness, where objective functions are combined in a weighted sum. A potential extension to current approaches would be to design more appropriate methods, without transforming the multiobjective problem into a single-objective one. Another interesting direction would be to use fitness landscape characterisation metrics as feedback strategies for both single- and multiobjective problems. Furthermore, feedback strategies for controlling the replacement strategy is a gap that could be addressed with future research.

5.2. Effect Assessment

The summary of effect assessment strategies shown in Table II indicates that most approaches use the change in the properties of the current solution with respect to the best solution (55 papers, more than 34% of all papers). This is followed by methods that use the properties of the current solution directly as an indication of the effect that parameters have on the performance of the search algorithm (47 papers, more than 29%). In the third and fourth position in terms of the total number of papers published, we have methods that consider the effect in the properties of the current solution with respect to the parent (i.e., ancestor), and with respect to the whole population (or a subset of solutions from the population). There are only two papers that consider the median solution when measuring the effect of parameter values [Julstrom 1995, 1997].

Figure 3(b) shows the number of papers over the years in the different effect assessment categories. The number of methods that use the ancestor(s) when estimating the effect of parameter values has increased steadily over the years. The lowest increase rate can be observed for methods that use the median and worst solution.

Davis [1991] made one of the first attempts to assess the effect of different crossover operators on the performance of a genetic algorithm. At every iteration, several crossover rates are applied simultaneously to create new solutions. The effect of each crossover operator is approximated using the *local delta*, denoted as δ_i , which estimates the difference in fitness gain of the offspring with respect to the best solution in the population. The rate of each crossover operator $\hat{c}_t(\hat{c}_i)$ at timestep t is calculated as $\hat{c}_t(\hat{c}_i) = 0.85\hat{c}_{t-1}(\hat{c}_i) + \delta_i$, where 0.85 is the weight given to the value of the crossover rate from the previous iteration. The parameter effect assessment mechanism used in this approach considers successful parameters as the ones that produce solutions whose fitnesses is close to that of the global best solution.

Rank proportional mutation [Sewell et al. 2005] is an extension to this work. The algorithm uses equal value intervals in the range $[\hat{m}_{r_{\min}}, \hat{m}_{r_{\max}}]$ for the mutation rate to sample from Cervantes and Stephens [2009]. The solutions are ranked according to the fitness function, and the mutation rate is selected according to these ranks as $\hat{m}_r(s) = \hat{m}_{r_{\min}} + (\hat{m}_{r_{\max}} - \hat{m}_{r_{\min}}) \cdot \frac{r(s)-1}{\mu-1}$, where $\hat{m}_r(s)$ refers to the mutation rate applied to solution s , $r(s)$ is the rank of solution s , and μ is the size of the population. The worse an individual's rank, the more drastic the increase in the mutation rate. As a result, the genes of low-fitness individuals have a lower probability of being propagated to the next iteration since there is a higher chance that they mutate as their viability declines.

The effect assessment strategy in *rank proportional mutation* considers the relative effect of parameter values. Parameter values compete with each other to produce solutions that are better ranked and have a better chance to be selected in the next iteration. In this respect, the *rank proportional mutation* is different from the approach using *local delta*, in which parameters do not have to compete with each other. In *local delta*, a parameter's selection rate depends only on the globally best solution; the aim is to produce solutions that are as close as possible to this best solution. In some sense, this creates an implicit competition with the parameter value that created the best solution.

A similar approach [Srinivas and Patnaik 1994] calculates the effect of mutation and crossover rates based on the best fitness $f(s^*)$ and the average fitness $\bar{f}$ of the population. The crossover rate $\hat{c}_r$ and mutation rate $\hat{m}_r$ are computed as follows:

$$\hat{c}_r = \frac{k_1(f(s^*) - \Delta f(s_{\hat{c}_r}))}{f(s^*) - \bar{f}}, \quad \hat{m}_r = \frac{k_2(f(s^*) - \Delta f(s_{\hat{m}_r}))}{f(s^*) - \bar{f}}, \quad (2)$$

where k_1 and k_2 are two constants selected in the range $[0.0, 1.0]$ and $\Delta f(s_{\hat{c}_r})$ $\Delta f(s_{\hat{m}_r})$ are the fitness gain of solutions created by applying c_r and m_r compared to their parents. The operator rates are inversely proportional to the difference between the best and the average fitness of the population. The operator rates increase if the difference between the best fitness and the fitness of the solution under consideration is low, that is, lower values of operator rates are applied to high-fitness solutions, and higher values of operator rates are used to change low-fitness solutions. The denominator represents the population as a whole; if the mean fitness approaches the maximum fitness, we can conclude that the search is stagnating and the operator rates have to increase to introduce more diversity into the population.

The Population Resizing and Fitness Improvement Genetic Algorithm (PRoFIGA) [Eiben et al. 2004] adjusts the population size based on the fitness improvement of the best solution in the population. The population grows if the best fitness in the population increases or if the best fitness does not change for a specific number of iterations. The population growth rate $\Delta\mu$ is computed as

$$\Delta\mu = \alpha(eval_{\max} - eval_t) \frac{f_t(s^*) - f_{t-1}(s^*)}{f_1(s^*)}, \quad (3)$$

where α is a parameter selected from the interval $(0,1)$, $eval_{\max}$ and $eval_t$ are the maximum number of fitness evaluations and the current evaluation number (timestep t), $f_t(s^*)$, $f_{t-1}(s^*)$, and $f_1(s^*)$ denote the current (timestep t), the previous (timestep $t - 1$) and the initial best-fitness values (timestep 1). If the best-fitness value does not improve in a certain number of evaluations, the authors propose to increase the population size. The motivation for varying the population size is to use a big population when exploration is desirable and a small population for exploiting the high-quality solutions. The experimental study shows that adapting the population size is beneficial for

the performance of the algorithm. Nevertheless, there is general agreement about the difficulty of adjusting the population size, since it interacts in complex ways with other parameters of an EA, such as the parent selection and replacement procedures.

Giger et al. [2007] employ the best-fitness frequency. The solutions are initially divided into classes according to their fitness values. Each class contains solutions with fitness values between $[f_{min}, f_{min} + \epsilon]$, where ϵ is the range size of the fitness class. The fitness classes are employed to compute the best-fitness frequency $\text{bff}_t(s_i)$ at every iteration t and for each solution s_i as follows:

$$\text{bff}_t(s_i) = \begin{cases} 1 & \text{if } |f(s_i) - f(s^*)| \leq \epsilon f(s^*) \\ 0 & \text{otherwise,} \end{cases} \quad (4)$$

where $f(s_i)$ is the fitness of a solution s_i and $f(s^*)$ is the fitness of the best solution in the population. The total bff_t of the current population is calculated as the average of the best-fitness values of all solutions, given by $\text{bff}_t = \frac{1}{N} \sum_i \text{bff}_t(s_i)$.

A low value of the bff measure indicates that the fitnesses of the solutions are very similar. This could mean that the search has stagnated in a local or global optimum, or that the fitness space is flat. On the other hand, a high bff value indicates that the fitness landscape is characterised by one or more steep unimodal or multimodal peaks, in which it is very likely to improve the quality of the solutions as the search progresses. The adaptation rate δ is calculated as

$$\delta = \begin{cases} -\frac{\delta_{max}}{th}(\text{bff} - th) & \text{if } \text{bff} \leq th \\ \frac{\delta_{max}}{th}(\text{bff} - th) & \text{otherwise.} \end{cases} \quad (5)$$

Parameter δ_{max} and the threshold value th control the balance between exploration and exploitation of parameter values. Best-fitness frequency does not only consider how the solutions improve in terms of fitness, but also how their fitnesses are distributed, that is, diversity of the population.

One of the main challenges in designing successful effect assessment frameworks is the choice of reference point. For instance, how should quality improvement be measured? Should it be based on the difference in quality of the current solution compared to its parent, or should it consider the best solution produced? A systematic investigation of appropriate effect assessment strategies would be valuable, and help design better adaptive parameter control methods.

5.3. Quality Attribution

Parameter quality refers to a scalar value attributed to a parameter value to describe its performance based on current or past observations. Table II shows that the majority of the approaches build a model to predict the quality of parameter values (around 42% of the papers are classified into the ‘learned’ quality attribution category). Around 34% of the approaches (55 papers) use the immediate effect as quality. The two last categories (average and extreme), contain the lowest percentage of papers (around 13% and 10%, respectively). Figure 3(c) shows how these numbers have evolved over time for the different subcategories.

Some parameter quality attribution methods use a predefined time window for which the effect of the parameters is recorded to approximate the quality of the parameter values used during the iterations covered by the time window. Igel et al. [2005] use this time window (adaptation cycle) to estimate the quality of parameter values as the average effect on algorithm performance. The PRoFIGA [Eiben et al. 2004], on the other hand, is an extreme quality attribution strategy since it focuses on the best-fitness values and ignores the other changes in the population.

Fialho et al. [2008, 2009, 2010] use an extreme quality attribution strategy, with a normalisation scheme that divides parameter effects by the best effect achieved by any parameter value. Whitacre et al. [2006] employ statistical tests to detect outliers. A parameter value is considered an outlier if it produces a solution that is statistically unlikely to occur. The parameter effect assessment strategy calculates the probability that a solution is an outlier, and the sum of all outlier probabilities of the solutions produced by a parameter value represents its quality. The goal is to reward parameter values that generate solutions that are rare but very fit compared to the other members of the population. Other parameter values that generate frequent solutions with small improvements are not regarded as important, thus are not used in the next iteration.

The Genetic Algorithm with Varying Population Size (GAVaPs) [Arabas et al. 1994] attributes parameter quality in a more indirect way, by assigning a lifetime to each solution when it is created. The lifetime variable is reduced at every iteration until it reaches zero, and the solution is removed from the population. The initial calculation of an individual's lifetime ($t(s)$) is influenced by the individual's fitness. Equation (6) depicts the three alternative lifetime calculations explored. In the first equation, t_p represents the simple fitness-proportionate allocation. The linear allocation t_l considers the fitness of the solution $f(s)$ in comparison with the best $f(s^*)$ and the worst fitnesses $f(s^-)$ in the population. The bilinear allocation t_b comprises a trade-off between proportional and linear allocation, defined as

$$\begin{aligned} t_p(s) &= \min \left(t_l + \frac{(t_u - t_l)f(s)}{2\bar{f}}, t_u \right) \\ t_l(s) &= t_l + \frac{(t_u - t_l)(f(s_i) - f(s^-))}{f(s^*) - f(s^-)} \\ t_b(s) &= \begin{cases} t_l + \frac{(t_u - t_l)(f(s) - f(s^-))}{2(\bar{f} - f(s^-))} & \text{if } \bar{f} \geq f(s) \\ \frac{t_l + t_u}{2} + \frac{(t_u - t_l)(f(s) - \bar{f})}{2(f(s^*) - \bar{f})} & \text{otherwise,} \end{cases} \end{aligned} \quad (6)$$

where lt_u and lt_l represent the upper and lower bounds of the lifetime for a solution, and $\bar{f}$ is the average fitness of the population. Parameters lt_u and lt_l require tuning; thus, to some extent, they defeat the purpose of making population size adaptive. The experimental studies on GAVaPs show that the linear strategy has the best performance but is the most computationally expensive [Arabas et al. 1994]. The bilinear allocation requires the least computations, but its performance is the worst. The authors argue that the proportional strategy is the best choice for its moderate performance and cost. Introduced as an extension of GAVaPs, the Adaptive Population size Genetic Algorithm (APGA) [Bäck et al. 2000] is a steady-state GA with the best solution having an unlimited lifetime. However, as explained in Smorodkina and Tauritz [2007a], APGA converges to $c/2(lt_u + lt_l)$, where c is the number of children generated per generation; thus, APGA does not truly adapt the population size.

One of the important issues in parameter control is the dependency between parameters. To tackle this problem, Bielza et al. [2013] used a Bayesian Network that models the interaction between different parameters by representing each parameter as a node of the network. The Bayesian Network is learned from the best parameter combinations, and used to generate new solutions in the next iterations.

The performance of EAs is highly susceptible to noise and error, involving random effects of parameter values. Thus, the discovery of high-quality solutions may not always be related to algorithm parameters. To overcome this limitation, Maturana and Saubion [2008b] introduced a fuzzy model that learns the effects of parameter values. After the learning phase, the model is employed to control parameter values during

the optimisation process. This is a first example that deals with noise in the effect of parameter values, and opens up interesting new research directions into assessing uncertainty in adaptive parameter control.

In general, adaptive parameter control methods derive the quality of a parameter value from recent, but nonetheless past, performance (time t). Rather than assuming that this value is representative of the parameter value's performance at time $t - 1$, it is possible to model and predict the performance. Such approaches might be grouped into a separate category were it not for their scarcity. Only Predictive Quality Attribution (PQA) [Aleti and Moser 2011; Aleti et al. 2014] uses time series forecasting to predict the performance of parameter values based on past performance. PQA monitors the effect of parameter value for each iteration. Instead of translating the effect in the last iteration(s) $t - 1$ directly into the quality for the parameter values, PQA applies time series prediction as a method of estimating the quality (time t) as part of the adaptive parameter control strategy.

A main challenge in quality attribution that has not been addressed by current research in a systematic way is the interaction between different parameters. The quality of one algorithm parameter depends on other parameter settings. Methods that model this interaction would be of high value to adaptive parameter control.

5.4. Parameter Update

The aim of parameter update is to decide whether to use parameter values that have proven successful in previous iterations, or to explore new value combinations. The majority of parameter update methods found in the literature are greedy strategies (60 papers, around 37% of all papers), for which the parameter value with the best quality is selected. The second most frequently studied parameter update strategy found in the literature is quality proportionate update, also known as probability matching (54 papers, around 33% of all papers, as indicated in Table II), in which the probability of applying a certain parameter value is proportional to the quality of that parameter value.

One such method [Corne et al. 2002] adapts the mutation rate using a quality proportionate update mechanism based on usage frequency and the fitness distance correlation. The update probability for each parameter value at timestep t is as follows:

$$p_t(v_{ij}) = \begin{cases} \frac{w}{\sum_{r=1}^m q_t(v_{ir})} \sum_{r=t-\tau}^{t-1} q_r(v_{ij}) + (1-w)p_{t-1}(v_{ij}), & \text{if } \sum_{s=1}^m q_t(v_{is}) > 0 \\ \frac{w}{m} + (1-w)p_{t-1}(v_{ij}), & \text{otherwise,} \end{cases} \quad (7)$$

where m is the number of possible values for parameter v_i , τ is the time window considered to approximate the quality of a parameter value, and $w \in (0, 1]$ is the weight given to each parameter value. For $w = 1$ and $\tau = 1$, the algorithm is similar to the approach proposed by Hong et al. [2000]. Igel and Kreutz [2003] introduced a lower limit for the probability p_{min} for each parameter value to ensure that underperforming parameter values do not disappear during optimisation, since they may be beneficial in the later stages of the search. The probability of using a certain parameter value is equal to

$$p'_t(v_{ij}) = p_{min} + (1 - mp_{min}) \frac{p_t(v_{ij})}{\sum_{r=1}^m p_t(v_{ir})}. \quad (8)$$

Dividing the probability of each parameter value by the sum of probabilities of all possible parameter values normalises the values in the interval $[0, 1]$. In probability matching techniques, the values resulting from the reward allocation may not always

Table III. Parameter Control Methods Classified by the Validation Category

Problem type			Statistical analysis		
Single-objective	140	88.050%	Mean	59	37.107%
Multi-objective	16	10.063%	Standard	25	15.723%
Constraints	25	15.572%	Median	2	1.258%
Benchmark method			Boxplots	5	3.145%
Static	114	71.698%	Statistical test	7	4.403%
Deterministic	32	20.126%	Success rate	17	10.692%
Self-adaptive	9	5.660%	Computations	9	5.660%
Adaptive	35	22.013%	Best	55	34.591%
Analysis	61	38.365%	Worst	15	9.434%
			None	78	49.057%
Problem category					
Quadratic assignment	6	3.774%	Simulation	9	5.660%
Industrial	30	18.868%	Benchmark	87	54.717%
Scheduling	3	1.887%	Minimal deceptive	2	1.258%
Royal road	9	5.660%	Long path	2	1.258%
One-max	12	7.547%	Cutting stock	2	1.258%
Travelling salesperson	5	3.145%	Layout	1	0.629%
Graph coloring	2	1.258%	Machine learning	4	2.516%
Satisfiability	11	6.918%	Knapsack	4	2.516%

reflect the relative differences in algorithm performance. Even if certain parameter values perform much better than all others, the increase in their probability of being chosen in the next step is only marginal.

Adaptive Pursuit (AP) [Thierens 2005] rewards the best-performing parameter values j^* by increasing their probability by $p_t(v_{ij})$ as follows:

$$p_t(v_{ij}) = \begin{cases} p_{t-1}(v_{ij}) + \beta(p_{\max} - p_{t-1}(v_{ij})) & \text{if } j = j^* \\ p_{t-1}(v_{ij}) + \beta(p_{\min} - p_{t-1}(v_{ij})) & \text{otherwise,} \end{cases} \quad (9)$$

where $p_{\max} = 1 - (m - 1)p_{\min}$ and m is the number of parameter values. AP is a greedier strategy compared to Probability Matching. Increasing the probabilities of the best-parameter values ensures an appropriate difference in probabilities in accordance with the performance experienced.

Even though every parameter value is selected from time to time, in practice, the AP algorithm would have to wait for a number of iterations before realising that some new parameter value has become the best one. To address this problem, the Dynamic Multi-Armed Bandit (DMAB) [Fialho et al. 2009] was introduced, which completely recalculates the quality of parameter values when a change-detection test flags a change in the effects distribution. The Page-Hinkley (PH) change-detection test restarts the algorithm when the effect of the parameter values has changed. As a result, DMAB can quickly identify the new best-parameter value without being slowed down by old information.

Some of the parameter value update strategies discretise the choices beforehand, but most sample values from defined ranges. The algorithm performance is attributed to the entire range from which the parameter value was sampled. Ranges that are not adjusted during the process are likely suboptimal. Using narrow ranges increases the accuracy of the feedback mechanism, but leads to increased combinatorial complexity. Defining wider ranges increases the sampling inaccuracy, because a new sample may be far from the previous sample that incurred the positive feedback.

Adaptive Range Parameter Selection (ARPS) [Aleti et al. 2012] deals with this problem by adjusting the ranges of continuous parameter values as the optimisation process progresses. The algorithm starts with two ranges for each parameter interval. At every iteration, ARPS rewards the best-performing parameter ranges by dividing them into two new ranges and assigning each new range the same probability as the parent range, whereas the worst-performing ranges are merged with one of the neighbouring ranges.

In ARPS, due to the stochastic behaviour of EAs, high-performing ranges may be merged with less successful ranges into very large intervals, making it difficult for the parameter control method to re-establish small, promising regions within the range. This behaviour almost certainly affects the performance of the search. Therefore, an improved approach to maintaining parameter value intervals was the subject of the investigation in the Entropy-based Adaptive Range Parameter Control (EARPC) algorithm [Aleti and Moser 2013].

The EARPC algorithm consists of two major steps. In the first step, it uses k-means clustering to determine how best to subdivide the values into groups. The second step uses these clusters as a basis to define the range boundaries for the groups. EARPC clusters parameter values by performance and subdivides the intervals according to the clustering. The process is repeated at every iteration; therefore, disadvantaging a well-performing interval has no effect beyond the subsequent iteration.

5.5. Validation

The validation procedure varies greatly among the papers included in this SLR. The different validation approaches can be classified based on (i) the problem type used during the optimisation, (ii) the specific problem category, (iii) the benchmark method used for comparison, and (iv) the statistical analysis conducted. It was interesting to see that most approaches consider single-objective problems (around 88% of the papers), and not many papers considered constraints (15.572%).

The aim of adaptive parameter control is to improve the performance of the original algorithm. Thus, it is important that the algorithms enhanced with the adaptive strategies are compared with the original method or other parameter control strategies. The majority of the approaches use static parameter settings as a benchmark for comparison (114 papers, 71.698%), in which algorithms are tuned prior to the optimisation process. Around 38% of the papers perform an analysis of the behaviour of the parameter-control strategy, in terms of sensitivity to hyper-parameters (i.e., parameters of the control strategy), theoretical investigations, or performance in a wide range of problems. This kind of analysis provides more insight into the weaknesses and strengths of the adaptive strategies.

The range of problems used to validate the adaptive parameter-control strategies is quite wide, with established benchmark functions dominating the literature (around 54%). For a detailed description of these problems, we refer the interested reader to Ausiello et al. [2012]. Well-known problems with established characteristics are seen to lend credibility to the experimentation. Interestingly, the second-highest number of papers uses industrial problems, whose relevance at least in the frame of the pertinent industry is beyond question. Using a problem generator permits shaping the problem to specific needs and varying difficulty, which provides opportunities to investigate algorithm performance more comprehensively.

Due to the stochastic behaviour of EAs, results concerning their performance are usually reported as mean values over repeated trials (around 37% of papers). For a fair comparison, the same number of function evaluations for each trial should be allocated [Rardin and Uzsoy 2001]. Many papers, however, did not present any comparison with the state of the art (78 papers, around 49%).

6. CROSS-ANALYSIS AND RESEARCH GAPS

The cross-analysis of the literature in terms of the different categories of taxonomy is performed in order to identify current gaps in the literature and potential topics for future research. We categorised and cross-analysed all papers in terms of algorithm parameters versus the different categories of parameter-control strategies. The results are shown as number of citations (in brackets) and relevant references in Tables IV and V.

6.1. Feedback Strategy versus Parameters Controlled

Independent of which parameter is being controlled, the information regarding the phenotype of the solutions is the most frequently researched feedback strategy. Interestingly, the phenotype-feedback strategy is more preferred than the feasibility feedback strategy, even when the fitness function is being controlled by adding a penalty for infeasibility. The numbers of papers in these two categories, however, are quite close; 24 papers employ a phenotype-feedback strategy, and 23 papers use feasibility.

Most papers that use phenotype feedback focus on the mutation rate, and very few of them consider the replacement strategy. The replacement strategy is not as frequently researched as the other algorithm parameters, with only one paper in the genotype feedback, and no papers in the other categories. This might be an interesting future direction, especially since the replacement strategy greatly affects the diversity of the population, and requires careful consideration.

The adaptation of solution representation is another interesting research direction. We could not find any papers that employ genotype, relative phenotype feedback, computations, and feasibility feedback in combination with this parameter. Solution representation is arguably the most difficult parameter to control and the choices are limited by the problem at hand. Nonetheless, since it affects the shape of the fitness landscape itself, it is an area that deserves some attention.

In highly constrained problems, finding feasible solutions is hard, and disruptive parameters that make drastic changes to solutions are not desirable. Thus, the feasibility of solutions is a useful feedback strategy in the update of parameter values that create solutions that satisfy constraints. Consequently, most papers that use this feedback strategy control the fitness function. We could not find any papers that employ solution feasibility in combination with crossover rate, solution representation, parent selection, replacement, and problem-specific parameters.

6.2. Effect Assessment Strategy versus Parameters Controlled

The effect of algorithm parameters on solution properties with respect to their ancestor (or parent) has mainly been studied in the context of controlling the genetic operators, such as crossover operator (12 papers), crossover rate (12 papers), mutation operator (15 papers), and mutation rate (10 papers). The majority of papers that study problem-specific parameters use ancestor-effect assessment (16 papers).

Naturally, in papers in which the population size is being adapted, population-effect assessment is the prevalent strategy (37 papers). Population has been observed to have a crucial effect on algorithm performance. The ideal population size depends on problem size and difficulty [Lobo and Lima 2005]. If the problem is simple and unimodal, the population size (μ) should be relatively small, as little diversity is required during the optimisation process. The algorithm would find the optimum most quickly through local optimisation. One might argue that if the problem is unimodal, the EA is preferably replaced with a simple local search. This assumes, however, that the fitness landscape of the problem is known or predictable.

Table IV. Feedback versus Parameters Controlled

Phenotype	Crossover operator (18) [Herrera and Lozano 2001; Hong et al. 2002; Pettinger and Everson 2002; Tongchhim and Chongstitvatana 2002; Luchian and Gheorghies 2003; Chen et al. 2005; Whitacre et al. 2006; Law and Szeto 2007; Maturana and Saubion 2008a; Maturana et al. 2009, 2010; Krempser et al. 2012; Aleli et al. 2014; Barkaoui and Gendreau 2013; Bielza et al. 2013; Duarte-Mermoud et al. 2013; Karafotias et al. 2014a; Li et al. 2014a]
	Crossover rate (35) [Grefenstette 1986; Davis 1989, 1991; Lee and Takagi 1993; Srinivas and Patnaik 1994; Julstrom 1995; Lis and Lis 1996; Julstrom 1997; Tuson and Ross 1998; Ho et al. 1999; Mak et al. 2000; Hong et al. 2002; Tongchhim and Chongstitvatana 2002; Luchian and Gheorghies 2003; Wong et al. 2003; Maeda et al. 2006; Eiben et al. 2007b; Giger et al. 2007; San-José-Revuelta 2007; Vafae et al. 2007; Wang et al. 2009b; Xu et al. 2010; Aleli and Moser 2011; McGinley et al. 2011; Nadi and Khader 2011; Aleli et al. 2012; Aleli and Moser 2013; Aleli et al. 2014; Bielza et al. 2013; Riff and Montero 2013; Karafotias et al. 2014a; Bhagwan Hanchate and S Bichkar 2014; Aleli 2014; Aleli and Grunске 2015; Aine et al. 2009]
	Mutation operator (27) [Budin et al. 1998; Hong et al. 2000; Herrera and Lozano 2001; Hong et al. 2002; Pettinger and Everson 2002; Tongchhim and Chongstitvatana 2002; Basseur et al. 2003; Luchian and Gheorghies 2003; Chen et al. 2005; Igel et al. 2005; Whitacre et al. 2006; Maturana and Saubion 2007; Fialho et al. 2008; Gomes et al. 2008; Maturana and Saubion 2008a; Vafae and Nelson 2010; Candan et al. 2012; Karafotias et al. 2012; Krempser et al. 2012; Aleli et al. 2014; Barkaoui and Gendreau 2013; Bielza et al. 2013; Candan et al. 2013; Maturana et al. 2012; Karafotias et al. 2014a; Li et al. 2014a, 2014b]
	Mutation rate (44) [Grefenstette 1986; Lee and Takagi 1993; Arnone et al. 1994; Srinivas and Patnaik 1994; Julstrom 1995; Lis 1996; Lis and Lis 1996; Julstrom 1997; Tuson and Ross 1998; Ho et al. 1999; Mak et al. 2000; Corne et al. 2002; Hong et al. 2002; Muller et al. 2002; Tongchhim and Chongstitvatana 2002; Luchian and Gheorghies 2003; Wong et al. 2003; Arnold 2005; Sewell et al. 2005; Maeda et al. 2006; Eiben et al. 2007b; Giger et al. 2007; San-José-Revuelta 2007; Vafae et al. 2007; Maturana and Saubion 2008b; Cervantes and Stephens 2009; Vafae and Nelson 2009; Wang et al. 2009b; Böttcher et al. 2010; Xu et al. 2010; Aleli and Moser 2011; McGinley et al. 2011; Nadi and Khader 2011; Aleli et al. 2012; Aleli and Moser 2013; Aleli et al. 2014; Bielza et al. 2013; De Giovanni et al. 2013; Riff and Montero 2013; Karafotias et al. 2014a; Bhagwan Hanchate and S Bichkar 2014; Aleli 2014; Aleli and Grunске 2015; Aine et al. 2009]
	Representation (3) [Arnone et al. 1994; Dumitrescu et al. 2001; Grosan and Oltean 2003]
	Problem specific (28) [Grefenstette 1986; Arabas et al. 1994; Dirk Schlierkamp-Voosen 1994; Tuson and Ross 1998; Bäck et al. 2000; Corne et al. 2002; Igel and Kreutz 2003; Wong et al. 2003; Eiben et al. 2004; Thierens 2005; Whitacre et al. 2006; Giger et al. 2007; Thierens 2007; Fialho et al. 2008, 2009; Maturana et al. 2009; Vafae and Nelson 2009; Fialho et al. 2010; Aleli and Moser 2011; Aleli et al. 2012; Krempser et al. 2012; Liu et al. 2012; Aleli and Moser 2013; Aleli et al. 2014; Khedkar and Subbaraman 2013; Liu et al. 2013; Li et al. 2014b; Aleli 2014]
	Population (31) [Grefenstette 1986; Goldberg et al. 1991; Lee and Takagi 1993; Arabas et al. 1994; Mathias and Whitley 1994; Dirk Schlierkamp-Voosen 1994; Lis and Lis 1996; Schlierkamp-Voosen and Mühlenbein 1996; Bäck et al. 2000; Fernandes et al. 2000; Fernandes and Rosa 2001; Iorio and Li 2002; Yu et al. 2003; Eiben et al. 2004; Yu et al. 2005; Eiben et al. 2007b; Smorodkina and Tauritz 2007a; Yu et al. 2007; Lardeux and Goëffon 2010; Montero and Riff 2011; Zhan and Zhang 2011; Ahari and Shariat-Panahi 2015; Aleli et al. 2014; Bielza et al. 2013; Riff and Montero 2013; Goldberg et al. 2014; Karafotias et al. 2014a; Trunfio 2015; Cook and Tauritz 2010; Goldman and Punch 2014]
	Parent selection (11) [Grefenstette 1986; Goldberg 1990; Budin et al. 1998; Mahnig and Mühlenbein 2001; Eiben et al. 2007b; Law and Szeto 2007; Smorodkina and Tauritz 2007b; Liang and Leung 2011; McGinley et al. 2011; Barkaoui and Gendreau 2013; Karami and Hasanadeh 2015]
	Fitness function (24) [Sakanashi et al. 1994; Coit et al. 1996; Rasheed 1998; Eggermont et al. 1999; Eggermont and Van Hemert 2001; Barbosa and Lemonge 2002; Ben Hamida and Schoenauer 2002; Barbosa and Lemonge 2003; Wagner et al. 2005; Tessema and Yen 2006; Wagner et al. 2007; Barbosa and Lemonge 2008; Wagner and Michalewicz 2008; Rocha and Fernandes 2009; Tessema and Yen 2009; Gan et al. 2010; Afanasyeva and Buzdalov 2011; Beaser et al. 2011; Buzdalova and Buzdalov 2012; Buzdalov and Buzdalova 2013; De Giovanni et al. 2013; Lin 2013; Montemurro et al. 2013; Vincenti et al. 2010]
	Replacement (2) [Maturana and Saubion 2008b; Bielza et al. 2013]
Genotype	Offspring size (3) [Hansen et al. 1995; Jansen et al. 2005; Nwamba and Tauritz 2009]
	Crossover operator (6) [Herrera and Lozano 1996, 2001; Law and Szeto 2007; Maturana and Saubion 2008a; Maturana et al. 2009, 2010]
	Crossover rate (4) [Herrera and Lozano 1996; Wong et al. 2003; Giger et al. 2007; McGinley et al. 2011]
	Mutation operator (7) [Budin et al. 1998; Herrera and Lozano 2001; Maturana and Saubion 2007, 2008a; Vafae and Nelson 2010; Karafotias et al. 2012; Maturana et al. 2012]
	Mutation rate (5) [Corne et al. 2002; Wong et al. 2003; Giger et al. 2007; Maturana and Saubion 2008b; McGinley et al. 2011]
	Representation (0)
	Problem specific (6) [Corne et al. 2002; Wong et al. 2003; Giger et al. 2007; Maturana et al. 2009; Liu et al. 2012, 2013]
	Population (3) [Smith 1993; Mathias and Whitley 1994; Lardeux and Goëffon 2010]
	Parent selection (8) [Smith 1993; Herrera and Lozano 1996; Budin et al. 1998; Law and Szeto 2007; Vajda et al. 2008; Liang and Leung 2011; McGinley et al. 2011; Guntly and Tauritz 2011]
	Fitness function (0) Replacement (1) [Maturana and Saubion 2008b]
Relative phenotype	Offspring size (0)
	Crossover operator (2) [Herrera and Lozano 1996; Sakurai et al. 2010]
	Crossover rate (3) [Herrera and Lozano 1996; San-José-Revuelta 2007; Aleli and Moser 2013]
	Mutation operator (1) [Sakurai et al. 2010]
	Mutation rate (2) [San-José-Revuelta 2007; Aleli and Moser 2013]
	Representation (0)
	Problem specific (1) [Aleli and Moser 2013]
	Population (2) [Tan et al. 2001; Yen and Lu 2002]
	Parent selection (2) [Herrera and Lozano 1996; Vajda et al. 2008]
	Fitness function (0) Replacement (0) Offspring size (0)
Feasibility	Genotype diversity
	Crossover operator (1) [Tan et al. 2009]
	Crossover rate (1) [Aine et al. 2009]
	Mutation operator (1) [Tan et al. 2009]
	Mutation rate (2) [Arnone et al. 1994; Aine et al. 2009]
	Representation (1) [Arnone et al. 1994]
	Problem specific (0)
	Population (6) [Smith and Smuda 1995; Harik et al. 1999; Wagner and Michalewicz 2001; Yu et al. 2003, 2005, 2007]
	Parent selection (0)
	Fitness function (0) Replacement (0) Offspring size (1) [Nwamba and Tauritz 2009]
Feasibility	Crossover operator (1) [Whitacre et al. 2006]
	Crossover rate (0) Mutation operator (1) [Whitacre et al. 2006]
	Mutation rate (1) [De Giovanni et al. 2013]
	Representation (0) Problem specific (0) Population (0) Selection (0)
	Fitness function (23) [Coit et al. 1996; Eiben et al. 1998; Kazarlis and Petridis 1998; Rasheed 1998; Eiben and van Hemert 1999; Hamda and Schoenauer 2000; Hamida and Schoenauer 2000; Nanakorn and Meesomklin 2001; Barbosa and Lemonge 2002; Ben Hamida and Schoenauer 2002; Barbosa and Lemonge 2003; Lemonge and Barbosa 2004; Tessema and Yen 2006; Barbosa and Lemonge 2008; Rocha and Fernandes 2009; Tessema and Yen 2009; Wang et al. 2009a; Gan et al. 2010; Beaser et al. 2011; Costa et al. 2013; Lin 2013; Montemurro et al. 2013; Vincenti et al. 2010]
	Replacement (0) Offspring size (0)

Table V. Effect Assessment versus Parameters Controlled

Ancestor	Crossover operator (12) [Herrera and Lozano 2001; Hong et al. 2002; Pettinger and Everson 2002; Luchian and Gheorghies 2003; Chen et al. 2005; Whitacre et al. 2006; Maturana et al. 2009; Sakurai et al. 2010; Krempser et al. 2012; Aleli et al. 2014; Duarte-Mermoud et al. 2013; Li et al. 2014a]	
	Crossover rate (12) [Davis 1989, 1991; Tuson and Ross 1998; Mak et al. 2000; Hong et al. 2002; Luchian and Gheorghies 2003; Vafae et al. 2007; Aleli and Moser 2011; McGinley et al. 2011; Aleli et al. 2012, 2014; Aleli 2014]	
	Mutation operator (15) [Hong et al. 2000; Herrera and Lozano 2001; Hong et al. 2002; Pettinger and Everson 2002; Luchian and Gheorghies 2003; Chen et al. 2005; Igel et al. 2005; Whitacre et al. 2006; Fialho et al. 2008; Sakurai et al. 2010; Krempser et al. 2012; Aleli et al. 2014; Maturana et al. 2012; Li et al. 2014a, 2014b]	
	Mutation rate (10) [Tuson and Ross 1998; Mak et al. 2000; Hong et al. 2002; Luchian and Gheorghies 2003; Vafae et al. 2007; Aleli and Moser 2011; McGinley et al. 2011; Aleli et al. 2012, 2014; Aleli 2014]	
	Representation (0)	
	Problem specific (16) [Tuson and Ross 1998; Igel and Kreutz 2003; Thierens 2005; Whitacre et al. 2006; Fialho et al. 2008, 2009; Maturana et al. 2009; Fialho et al. 2010; Aleli and Moser 2011; Aleli et al. 2012; Krempser et al. 2012; Liu et al. 2012; Aleli et al. 2014; Liu et al. 2013; Li et al. 2014b; Aleli 2014]	
	Population (2) [Lardeux and Geoffon 2010; Aleli et al. 2014]	
	Parent selection (3) [Goldberg 1990; McGinley et al. 2011; Guntly and Tauritz 2011]	
	Fitness function (3) [Buzdalova and Buzdalov 2012; Buzdalov and Buzdalova 2013; Lin 2013]	
	Replacement (0)	
Population	Offspring size (2) [Hansen et al. 1995; Jansen et al. 2005]	
	Crossover operator (3) [Aleli et al. 2014; Bielza et al. 2013; Karafotias et al. 2014a]	
	Crossover rate (8) [Grefenstette 1986; Lee and Takagi 1993; Lis and Lis 1996; Eiben et al. 2007b; Aleli et al. 2014; Bielza et al. 2013; Riff and Montero 2013; Karafotias et al. 2014a]	
	Mutation operator (3) [Aleli et al. 2014; Bielza et al. 2013; Karafotias et al. 2014a]	
	Mutation rate (8) [Grefenstette 1986; Lee and Takagi 1993; Lis and Lis 1996; Eiben et al. 2007b; Aleli et al. 2014; Bielza et al. 2013; Riff and Montero 2013; Karafotias et al. 2014a]	
	Representation (0)	
	Problem specific (6) [Grefenstette 1986; Arabas et al. 1994; Dirk Schlierkamp-Voosen 1994; Bäck et al. 2000; Eiben et al. 2004; Aleli et al. 2014]	
	Population (37) [Grefenstette 1986; Goldberg et al. 1991; Lee and Takagi 1993; Smith 1993; Arabas et al. 1994; Mathias and Whitley 1994; Dirk Schlierkamp-Voosen 1994; Smith and Smuda 1995; Lis and Lis 1996; Schlierkamp-Voosen and Mühlenbein 1996; Harik et al. 1999; Bäck et al. 2000; Fernandes et al. 2000; Fernandes and Rosa 2001; Tan et al. 2001; Wagner and Michalewicz 2001; Iorio and Li 2002; Yen and Lu 2002; Yu et al. 2003; Eiben et al. 2004; Yu et al. 2005; Eiben et al. 2007b; Smorodkina and Tauritz 2007a; Yu et al. 2007; Lardeux and Geoffon 2010; Montero and Riff 2011; Zhan and Zhang 2011; Ahrari and Shariat-Panahi 2015; Aleli et al. 2014; Bielza et al. 2013; Riff and Montero 2013; Goldberg et al. 2014; Karafotias et al. 2014a; Trunfio 2015; Cook and Tauritz 2010; Goldman and Punch 2014]	
	Parent selection (3) [Grefenstette 1986; Smith 1993; Eiben et al. 2007b]	
	Fitness function (0)	
Best	Replacement (1) [Bielza et al. 2013]	
	Offspring size (0)	
	Crossover operator (2) [Luchian and Gheorghies 2003; Karafotias et al. 2014a]	
	Crossover rate (12) [Grefenstette 1986; Lee and Takagi 1993; Srinivas and Patnaik 1994; Lis and Lis 1996; Ho et al. 1999; Luchian and Gheorghies 2003; Maeda et al. 2006; Giger et al. 2007; Wang et al. 2010; Karafotias et al. 2014a; Bhagwan Hanchate and S Bichkar 2014]	
	Mutation operator (6) [Budin et al. 1998; Basseur et al. 2003; Luchian and Gheorghies 2003; Gomes et al. 2008; Karafotias et al. 2012, 2014a] [Grefenstette 1986; Lee and Takagi 1993; Arnone et al. 1994; Srinivas and Patnaik 1994; Lis and Lis 1996; Ho et al. 1999; Luchian and Gheorghies 2003; Sewell et al. 2005; Maeda et al. 2006; Giger et al. 2007; Vafae and Nelson 2009; Wang et al. 2009b; Xu et al. 2010; De Giovanni et al. 2013; Karafotias et al. 2014a; Bhagwan Hanchate and S Bichkar 2014]	
	Mutation rate (16) [Grefenstette 1986; Lee and Takagi 1993; Arnone et al. 1994; Srinivas and Patnaik 1994; Lis and Lis 1996; Ho et al. 1999; Luchian and Gheorghies 2003; Sewell et al. 2005; Maeda et al. 2006; Giger et al. 2007; Vafae and Nelson 2009; Wang et al. 2009b; Xu et al. 2010; De Giovanni et al. 2013; Karafotias et al. 2014a; Bhagwan Hanchate and S Bichkar 2014]	
	Representation (1) [Arnone et al. 1994]	
	Problem specific (8) [Grefenstette 1986; Arabas et al. 1994; Bäck et al. 2000; Igel and Kreutz 2003; Eiben et al. 2004; Giger et al. 2007; Vafae and Nelson 2009; Khedkar and Subbaraman 2013]	
	Population (17) [Grefenstette 1986; Lee and Takagi 1993; Arabas et al. 1994; Mathias and Whitley 1994; Lis and Lis 1996; Schlierkamp-Voosen and Mühlenbein 1996; Bäck et al. 2000; Fernandes et al. 2000; Fernandes and Rosa 2001; Iorio and Li 2002; Eiben et al. 2004; Smorodkina and Tauritz 2007a; Yu et al. 2007; Ahrari and Shariat-Panahi 2015; Karafotias et al. 2014a; Trunfio 2015; Cook and Tauritz 2010]	
	Parent selection (4) [Grefenstette 1986; Budin et al. 1998; Smorodkina and Tauritz 2007b; Vajda et al. 2008]	
Worst	Fitness function (19) [Coit et al. 1996; Rasheed 1998; Eggermont et al. 1999; Eggermont and Van Hemert 2001; Nanakorn and Meesomklin 2001; Barbosa and Lemonge 2003; Tessema and Yen 2006; Barbosa and Lemonge 2008; Rocha and Fernandes 2009; Tessema and Yen 2009; Wang et al. 2009a; Gan et al. 2010; Afanasyeva and Buzdalov 2011; Beaser et al. 2011; Costa et al. 2013; De Giovanni et al. 2013; Lin 2013; Montemurro et al. 2013; Vincenti et al. 2010]	
	Replacement (0) Offspring size (0)	
	Crossover operator (0)	
	Crossover rate (1) [Lee and Takagi 1993]	
	Mutation operator (1) [Budin et al. 1998]	
	Mutation rate (2) [Lee and Takagi 1993; Sewell et al. 2005]	
	Representation (0)	
	Problem specific (2) [Arabas et al. 1994; Bäck et al. 2000]	
	Population (6) [Lee and Takagi 1993; Arabas et al. 1994; Mathias and Whitley 1994; Bäck et al. 2000; Iorio and Li 2002; Cook and Tauritz 2010]	
	Parent selection (1) [Budin et al. 1998]	
Median	Fitness function (7) [Barbosa and Lemonge 2003; Tessema and Yen 2006; Barbosa and Lemonge 2008; Tessema and Yen 2009; Wang et al. 2009a; Gan et al. 2010; Lin 2013]	
	Replacement (0)	
	Offspring size (0)	
	Crossover operator (0)	
	Crossover rate (2) [Julstrom 1995, 1997]	
	Mutation operator (0)	
	Mutation rate (2) [Julstrom 1995, 1997]	
	Representation (0)	
	Problem specific (0)	
	Population (0)	
	Selection (0)	
	Fitness function (0)	
	Replacement (0)	
	Offspring size (0)	

(Continued)

Table V. Continued

Current	Crossover operator (8) [Herrera and Lozano 1996; Hong et al. 2002; Tongchim and Chongstitvatana 2002; Law and Szeto 2007; Maturana and Saubion 2008a; Tan et al. 2009; Barkaoui and Gendreau 2013; Bielza et al. 2013]
	Crossover rate (8) [Herrera and Lozano 1996; Hong et al. 2002; Tongchim and Chongstitvatana 2002; Wong et al. 2003; Eiben et al. 2007b; Wang et al. 2009b; Xu et al. 2010; Bielza et al. 2013]
	Mutation operator (9) [Hong et al. 2002; Tongchim and Chongstitvatana 2002; Maturana and Saubion 2007, 2008a; Tan et al. 2009; Candan et al. 2012; Barkaoui and Gendreau 2013; Bielza et al. 2013; Candan et al. 2013]
	Mutation rate (14) [Arnane et al. 1994; Corne et al. 2002; Hong et al. 2002; Muller et al. 2002; Tongchim and Chongstitvatana 2002; Wong et al. 2003; Arnold 2005; Eiben et al. 2007b; Maturana and Saubion 2008b; Cervantes and Stephens 2009; Wang et al. 2009b; Böttcher et al. 2010; Xu et al. 2010; Bielza et al. 2013]
	Representation (3) [Arnane et al. 1994; Dumitrescu et al. 2001; Grosan and Oltean 2003]
	Problem specific (4) [Dirk Schlierkamp-Voosen 1994; Corne et al. 2002; Wong et al. 2003; Thierens 2007]
	Population (11) [Dirk Schlierkamp-Voosen 1994; Fernandes et al. 2000; Fernandes and Rosa 2001; Wagner and Michalewicz 2001; Iorio and Li 2002; Eiben et al. 2007b; Smorodkina and Tauritz 2007a; Montero and Riff 2011; Zhan and Zhang 2011; Bielza et al. 2013; Cook and Tauritz 2010]
	Parent selection (7) [Goldberg 1990; Herrera and Lozano 1996; Eiben et al. 2007b; Law and Szeto 2007; Liang and Leung 2011; Barkaoui and Gendreau 2013; Guntly and Tauritz 2011]
	Fitness function (8) [Sakanashi et al. 1994; Kazarlis and Petridis 1998; Hamda and Schoenauer 2000; Nanakorn and Meesomklin 2001; Barbosa and Lemonge 2002; Lemonge and Barbosa 2004; Tessema and Yen 2006, 2009]
	Replacement (2) [Maturana and Saubion 2008b; Bielza et al. 2013] Offspring size (1) [Jansen et al. 2005]

Both best- and worst-effect assessment strategies are mainly used in combination with fitness function, which is controlled by adapting the penalty rate in constrained problems. Improvement in the worst solution might mean that new gradients are being explored, which indicates a better exploration of the search space. The median-effect strategy has been investigated in only two papers in the context of crossover and mutation rates, while strategies that focus on the current effect consider all algorithm parameters, with the mutation rate and population being studied in the majority of the papers (14 and 11 papers, respectively).

There is certainly a need for more investigation in the different effect assessment strategies in combination with replacement and parent selection procedures, for which papers are quite scarce compared to other parameters. The greedier the parent-selection strategy, the faster the algorithm converges. This, however, decreases the chance of finding the globally optimal solution(s), since the population is more prone to confinement in local optima. Thus, the parent-selection procedure should allow for the exploration of the search space, especially during the initial iterations, while focusing on the best solutions at the end. This suggests that, despite the relatively low number of papers that adjust this parameter online, an adaptation of the parent selection strategy could improve the performance of the algorithm. Effective adaptation of the parent-selection strategy may require alternative effect assessment strategies.

6.3. Quality Attribution Strategy versus Parameters Controlled

The classification of papers based on quality attribution strategy versus the algorithm parameters is depicted in Table VI. Immediate quality attribution is often used to control the mutation rate (18 papers), crossover rate (14 papers), fitness function (14 papers) and population (12 papers). The learned quality attribution has also been widely researched, with the majority of the papers focussing on population (17 papers), crossover rate (17 papers), and mutation operator (16 papers). Problem-specific operators have been controlled using a variety of effect assessment strategies, ranging from immediate (8 papers) to learned (10 papers). Replacement, on the other hand, is adjusted only with a learned quality-attribution strategy. Extreme and average quality-attribution strategies have the lowest number of papers, most of which focus on problem-specific parameters.

6.4. Parameter Update Strategy versus Parameters Controlled

The most frequently studied parameter update strategies are quality proportionate and greedy, which are at the two extremes of the exploration versus exploitation trade-off. The greedy methods focus mainly on the adaptation of the fitness function (16 papers),

Table VI. Quality Attribution versus Parameters Controlled

Immediate	Crossover operator (5) [Tongchim and Chongstitvatana 2002; Maturana and Saubion 2008a; Tan et al. 2009; Maturana et al. 2010; Barkaoui and Gendreau 2013]	
	Crossover rate (14) [Grefenstette 1986; Julstrom 1995, 1997; Mak et al. 2000; Tongchim and Chongstitvatana 2002; Wong et al. 2003; San-José-Revuelta 2007; Wang et al. 2009b; Xu et al. 2010; McGinley et al. 2011; Nadi and Khader 2011; Aleti et al. 2012; Aleti and Moser 2013; Riff and Montero 2013]	
	Mutation operator (9) [Budin et al. 1998; Tongchim and Chongstitvatana 2002; Basseur et al. 2003; Gomes et al. 2008; Maturana and Saubion 2008a; Tan et al. 2009; Karafotias et al. 2012; Barkaoui and Gendreau 2013; Maturana et al. 2012]	
	Mutation rate (18) [Grefenstette 1986; Arnone et al. 1994; Julstrom 1995, 1997; Mak et al. 2000; Tongchim and Chongstitvatana 2002; Wong et al. 2003; Sewell et al. 2005; San-José-Revuelta 2007; Wang et al. 2009b; Böttcher et al. 2010; Xu et al. 2010; McGinley et al. 2011; Nadi and Khader 2011; Aleti et al. 2012; Aleti and Moser 2013; De Giovanni et al. 2013; Riff and Montero 2013]	
	Representation (1) [Arnone et al. 1994]	
	Problem specific (8) [Grefenstette 1986; Arabas et al. 1994; Bäck et al. 2000; Wong et al. 2003; Eiben et al. 2004; Aleti et al. 2012; Aleti and Moser 2013; Khedkar and Subbaraman 2013]	
	Population (12) [Grefenstette 1986; Arabas et al. 1994; Mathias and Whitley 1994; Bäck et al. 2000; Fernandes et al. 2000; Fernandes and Rosa 2001; Wagner and Michalewicz 2001; Iorio and Li 2002; Eiben et al. 2004; Riff and Montero 2013; Trunfio 2015; Goldman and Punch 2014]	
	Parent selection (6) [Grefenstette 1986; Goldberg 1990; Budin et al. 1998; McGinley et al. 2011; Barkaoui and Gendreau 2013; Karami and Hasanzadeh 2015]	
	Fitness function (14) [Sakanashi et al. 1994; Coit et al. 1996; Kazarlis and Petridis 1998; Rasheed 1998; Eggermont et al. 1999; Hamda and Schoenauer 2000; Eggermont and Van Hemert 2001; Barbosa and Lemonge 2002; Ben Hamida and Schoenauer 2002; Wang et al. 2009a; Gan et al. 2010; De Giovanni et al. 2013; Lin 2013]	
	Replacement (0)	
Average	Offspring size (1) [Jansen et al. 2005]	
	Crossover operator (1) [Law and Szeto 2007]	
	Crossover rate (3) [Srinivas and Patnaik 1994; Lis and Lis 1996; Vafae et al. 2007]	
	Mutation operator (1) [Hong et al. 2000]	
	Mutation rate (5) [Srinivas and Patnaik 1994; Lis and Lis 1996; Corne et al. 2002; Vafae et al. 2007; Vafae and Nelson 2009]	
	Representation (0)	
	Problem specific (5) [Corne et al. 2002; Fialho et al. 2009; Vafae and Nelson 2009; Liu et al. 2012, 2013]	
	Population (4) [Lis and Lis 1996; Smorodkina and Tauritz 2007a; Yu et al. 2007; Lardeux and Geoffon 2010]	
	Selection (2) [Mahnig and Muhlenbein 2001; Law and Szeto 2007]	
	Fitness function (5) [Nanakorn and Meesomklin 2001; Barbosa and Lemonge 2002; Lemonge and Barbosa 2004; Buzdalova and Buzdalov 2012; Buzdalov and Buzdalova 2013]	
Learned	Replacement (0)	
	Offspring size (1) [Nwamba and Tauritz 2009]	
	Crossover operator (12) [Herrera and Lozano 1996; Hong et al. 2002; Pettinger and Everson 2002; Luchian and Gheorghies 2003; Chen et al. 2005; Sakurai et al. 2010; Krempser et al. 2012; Aleti et al. 2014; Bielza et al. 2013; Karafotias et al. 2014a; Li et al. 2014a]	
	Crossover rate (17) [Davis 1989, 1991; Lee and Takagi 1993; Herrera and Lozano 1996; Tuson and Ross 1998; Ho et al. 1999; Hong et al. 2002; Luchian and Gheorghies 2003; Maeda et al. 2006; Eiben et al. 2007b; Aleti and Moser 2011; Aleti et al. 2014; Bielza et al. 2013; Karafotias et al. 2014a; Bhagwan Hanchate and S Bichkar 2014; Aleti 2014; Aleti and Grunске 2015]	
	Mutation operator (16) [Hong et al. 2002; Pettinger and Everson 2002; Luchian and Gheorghies 2003; Chen et al. 2005; Igel et al. 2005; Maturana and Saubion 2007; Fialho et al. 2008; Sakurai et al. 2010; Karafotias et al. 2012; Krempser et al. 2012; Aleti et al. 2014; Bielza et al. 2013; Maturana et al. 2012; Karafotias et al. 2014a; Li et al. 2014a, 2014b]	
	Mutation rate (19) [Lee and Takagi 1993; Lis 1996; Tuson and Ross 1998; Ho et al. 1999; Hong et al. 2002; Muller et al. 2002; Luchian and Gheorghies 2003; Arnold 2005; Maeda et al. 2006; Eiben et al. 2007b; Maturana and Saubion 2008b; Cervantes and Stephens 2009; Aleti and Moser 2011; Aleti et al. 2014; Bielza et al. 2013; Karafotias et al. 2014a; Bhagwan Hanchate and S Bichkar 2014; Aleti 2014; Aleti and Grunске 2015]	
	Representation (2) [Dumitrescu et al. 2001; Grosan and Oltean 2003]	
	Problem specific (10) [Tuson and Ross 1998; Igel and Kreutz 2003; Thierens 2005; Fialho et al. 2008, 2010; Aleti and Moser 2011; Krempser et al. 2012; Aleti et al. 2014; Li et al. 2014b; Aleti 2014]	
	Population (17) [Goldberg et al. 1991; Lee and Takagi 1993; Smith 1993; Smith and Smuda 1995; Schlierkamp-Voosen and Mühlenbein 1996; Harik et al. 1999; Yu et al. 2003, 2005; Eiben et al. 2007b; Yu et al. 2007; Montero and Riff 2011; Zhan and Zhang 2011; Ahrari and Shariat-Panahi 2015; Aleti et al. 2014; Bielza et al. 2013; Goldberg et al. 2014; Karafotias et al. 2014a]	
	Selection (6) [Smith 1993; Herrera and Lozano 1996; Eiben et al. 2007b; Smorodkina and Tauritz 2007b; Vajda et al. 2008; Liang and Leung 2011]	
Extreme	Fitness function (11) [Barbosa and Lemonge 2003; Wagner et al. 2005; Tessema and Yen 2006; Wagner et al. 2007; Barbosa and Lemonge 2008; Wagner and Michalewicz 2008; Tessema and Yen 2009; Afanasieva and Buzdalov 2011; Beaser et al. 2011; Costa et al. 2013; Vincenti et al. 2010]	
	Replacement (2) [Maturana and Saubion 2008b; Bielza et al. 2013] Offspring size (0)	
	Crossover operator (2) [Whitacre et al. 2006; Maturana et al. 2009]	
	Crossover rate (3) [Giger et al. 2007; Vafae et al. 2007; Aine et al. 2009]	
	Mutation operator (4) [Whitacre et al. 2006; Vafae and Nelson 2010; Candan et al. 2012, 2013]	
	Mutation rate (3) [Giger et al. 2007; Vafae et al. 2007; Aine et al. 2009]	
	Representation (0)	
	Problem specific (6) [Dirk Schlierkamp-Voosen 1994; Whitacre et al. 2006; Giger et al. 2007; Thierens 2007; Fialho et al. 2009; Maturana et al. 2009]	
	Population (3) [Dirk Schlierkamp-Voosen 1994; Tan et al. 2001; Yen and Lu 2002]	
	Parent selection (0)	
	Fitness function (3) [Eiben et al. 1998; Eiben and van Hemert 1999; Montemurro et al. 2013]	
	Replacement (0)	
	Offspring size (1) [Hansen et al. 1995]	

whereas quality proportionate methods adjust mainly the mutation rate (16 papers). There is a need for further research in employing deterministic strategies to control the replacement function, for which we could not find any papers.

The replacement procedure affects the diversity of the population [Arabas et al. 1994]. A high replacement pressure decreases the diversity of the solutions and vice versa [Arabas et al. 1994]. Adapting this parameter might help improve on the premature convergence of the algorithm; thus, there is great potential in investigating different adaptive strategies for the replacement procedure. In general, apart from the

Table VII. Parameter Update versus Parameters Controlled

Quality proportionate (QP)	Crossover operator (12) [Hong et al. 2002; Pettinger and Everson 2002; Chen et al. 2005; Whitacre et al. 2006; Law and Szeto 2007; Maturana and Saubion 2008a; Maturana et al. 2010; Sakurai et al. 2010; Krempser et al. 2012; Barkaoui and Gendreau 2013; Bielza et al. 2013; Karafotias et al. 2014a]
	Crossover rate (13) [Grefenstette 1986; Davis 1989, 1991; Srinivas and Patnaik 1994; Tuson and Ross 1998; Hong et al. 2002; Giger et al. 2007; Vafaei et al. 2007; McGinley et al. 2011; Nadi and Khader 2011; Bielza et al. 2013; Riff and Montero 2013; Karafotias et al. 2014a]
	Mutation operator (15) [Hong et al. 2000, 2002; Pettinger and Everson 2002; Basseur et al. 2003; Chen et al. 2005; Igel et al. 2005; Whitacre et al. 2006; Gomes et al. 2008; Maturana and Saubion 2008a; Sakurai et al. 2010; Krempser et al. 2012; Barkaoui and Gendreau 2013; Bielza et al. 2013; Maturana et al. 2012; Karafotias et al. 2014a]
	Mutation rate (16) [Grefenstette 1986; Srinivas and Patnaik 1994; Tuson and Ross 1998; Hong et al. 2002; Muller et al. 2002; Sewell et al. 2005; Giger et al. 2007; Vafaei et al. 2007; Cervantes and Stephens 2009; Vafaei and Nelson 2009; Böttcher et al. 2010; McGinley et al. 2011; Nadi and Khader 2011; Bielza et al. 2013; Riff and Montero 2013; Karafotias et al. 2014a]
	Representation (0)
	Problem specific (12) [Grefenstette 1986; Arabas et al. 1994; Tuson and Ross 1998; Bäck et al. 2000; Igel and Kreutz 2003; Thierens 2005; Whitacre et al. 2006; Giger et al. 2007; Fialho et al. 2009; Vafaei and Nelson 2009; Fialho et al. 2010; Krempser et al. 2012]
	Population (15) [Grefenstette 1986; Goldberg et al. 1991; Smith 1993; Arabas et al. 1994; Smith and Smuda 1995; Bäck et al. 2000; Iorio and Li 2002; Yu et al. 2003, 2005; Montero and Riff 2011; Bielza et al. 2013; Riff and Montero 2013; Goldberg et al. 2014; Karafotias et al. 2014a; Cook and Tauritz 2010]
	Parent selection (6) [Grefenstette 1986; Smith 1993; Law and Szeto 2007; Liang and Leung 2011; McGinley et al. 2011; Barkaoui and Gendreau 2013]
	Fitness function (5) [Eiben et al. 1998; Kazarlis and Petridis 1998; Eiben and van Hemert 1999; Hamida and Schoenauer 2000; Lemonge and Barbosa 2004]
	Replacement (1) [Bielza et al. 2013] Offspring size (0)
QP with minimum probability	Crossover operator (6) [Luchian and Gheorghies 2003; Whitacre et al. 2006; Maturana and Saubion 2008a; Maturana et al. 2010; Aleli et al. 2014; Li et al. 2014a]
	Crossover rate (11) [Davis 1989, 1991; Julstrom 1995, 1997; Luchian and Gheorghies 2003; Aleli and Moser 2011; Aleli et al. 2012; Aleli and Moser 2013; Aleli et al. 2014; Aleli 2014; Aleli and Grunke 2015]
	Mutation operator (9) [Luchian and Gheorghies 2003; Igel et al. 2005; Whitacre et al. 2006; Gomes et al. 2008; Maturana and Saubion 2008a; Vafaei and Nelson 2010; Aleli et al. 2014; Li et al. 2014a, 2014b]
	Mutation rate (9) [Julstrom 1995, 1997; Luchian and Gheorghies 2003; Aleli and Moser 2011; Aleli et al. 2012; Aleli and Moser 2013; Aleli et al. 2014; Aleli 2014; Aleli and Grunke 2015]
	Representation (0)
	Problem specific (9) [Igel and Kreutz 2003; Whitacre et al. 2006; Thierens 2007; Aleli and Moser 2011; Aleli et al. 2012; Aleli and Moser 2013; Aleli et al. 2014; Li et al. 2014b; Aleli 2014]
	Population (1) [Aleli et al. 2014]
	Parent selection (0)
	Fitness function (0)
	Replacement (0) Offspring size (0)
Greedy	Crossover operator (5) [Herrera and Lozano 1996; Tongcham and Chongstitvatana 2002; Maturana et al. 2009; Tan et al. 2009; Krempser et al. 2012]
	Crossover rate (10) [Herrera and Lozano 1996; Lis and Lis 1996; Ho et al. 1999; Tongcham and Chongstitvatana 2002; Wong et al. 2003; Maeda et al. 2006; Eiben et al. 2007b; San-José-Revuelta 2007; Wang et al. 2009b; Xu et al. 2010]
	Mutation operator (9) [Budini et al. 1998; Tongcham and Chongstitvatana 2002; Maturana and Saubion 2007; Fialho et al. 2008; Tan et al. 2009; Candan et al. 2012; Karafotias et al. 2012; Krempser et al. 2012; Candan et al. 2013]
	Mutation rate (13) [Arnone et al. 1994; Lis and Lis 1996; Ho et al. 1999; Corne et al. 2002; Tongcham and Chongstitvatana 2002; Wong et al. 2003; Maeda et al. 2006; Eiben et al. 2007b; San-José-Revuelta 2007; Maturana and Saubion 2008b; Wang et al. 2009b; Xu et al. 2010; De Giovanni et al. 2013]
	Representation (1) [Arnone et al. 1994]
	Problem specific (12) [Dirk Schlierkamp-Voosen 1994; Corne et al. 2002; Wong et al. 2003; Eiben et al. 2004; Thierens 2005, 2007; Fialho et al. 2008; Maturana et al. 2009; Krempser et al. 2012; Liu et al. 2012; Khedkar and Subbaraman 2013; Liu et al. 2013]
	Population (16) [Dirk Schlierkamp-Voosen 1994; Lis and Lis 1996; Schlierkamp-Voosen and Mühlenbein 1996; Fernandes et al. 2000; Fernandes and Rosa 2001; Tan et al. 2001; Wagner and Michalewicz 2001; Yen and Lu 2002; Eiben et al. 2004, 2007b; Smorodkina and Tauritz 2007a; Yu et al. 2007; Lardeux and Geoffon 2010; Zhan and Zhang 2011; Trunfio 2015]
	Parent selection (6) [Goldberg 1990; Herrera and Lozano 1996; Budini et al. 1998; Eiben et al. 2007b; Smorodkina and Tauritz 2007b; Guntly and Tauritz 2011]
	Fitness function (16) [Sakanashi et al. 1994; Rasheed 1998; Eggermont et al. 1999; Eggermont and Van Hemert 2001; Wagner et al. 2005, 2007; Wagner and Michalewicz 2008; Rocha and Fernandes 2009; Wang et al. 2009a; Gan et al. 2010; Afanasyeva and Buzdalov 2011; Buzdalova and Buzdalov 2012; Buzdalov and Buzdalova 2013; De Giovanni et al. 2013; Montemurro et al. 2013]
	Replacement (1) [Maturana and Saubion 2008b]
Deterministic	Crossover operator (2) [Herrera and Lozano 2001; Duarte-Mermoud et al. 2013]
	Crossover rate (4) [Lee and Takagi 1993; Mak et al. 2000; Bhagwan Hanchate and S Bichkar 2014; Aine et al. 2009]
	Mutation operator (1) [Herrera and Lozano 2001]
	Mutation rate (6) [Lee and Takagi 1993; Lis 1996; Mak et al. 2000; Arnold 2005; Bhagwan Hanchate and S Bichkar 2014; Aine et al. 2009]
	Representation (2) [Dumitrescu et al. 2001; Grosan and Oltean 2003]
	Problem specific (0)
	Population (4) [Lee and Takagi 1993; Mathias and Whitley 1994; Ahrari and Shariat-Panahi 2015; Goldman and Punch 2014]
	Parent selection (3) [Mahning and Mühlenbein 2001; Vajda et al. 2008; Karami and Hasanazadeh 2015]
	Fitness function (12) [Coit et al. 1996; Hamda and Schoenauer 2000; Nanakorn and Meesomklin 2001; Barbosa and Lemonge 2002, 2003; Tessema and Yen 2006; Barbosa and Lemonge 2008; Tessema and Yen 2009; Beaser et al. 2011; Costa et al. 2013; Lin 2013; Vincenti et al. 2010]
	Replacement (0)
	Offspring size (2) [Hansen et al. 1995; Jansen et al. 2005]

fitness function, there are not many papers that employ the deterministic strategy to control other algorithm parameters, which is another potential area for investigation.

Solution representation has been investigated only in combination with deterministic and greedy strategies. One of the three papers that control solution representation controls it in combination with mutation rate [Arnone et al. 1994]; the other two control it in isolation [Grosan and Oltean 2003; Dumitrescu et al. 2001]. Arnone et al. [1994] address the problem of setting the maximum length in a variable-length genotype representation. Naturally, shorter genotypes would be faster to evaluate; however, the

optimal genotype length depends on the problem instance and possibly the optimisation stage. The genotype length is adjusted based on the average genotype length $\bar{l}$ and the length of the best solution l^* . The maximum length is increased if the ratio $l^*/\bar{l}$ is above a threshold, and decreased otherwise. There is no indication of appropriate settings for the threshold value, which would be an interesting area for investigation.

7. DISCUSSION AND RESEARCH CHALLENGES

The benefits of adaptive parameter control are obvious. If carried out sufficiently, adjusting parameters throughout the process based on feedback ensures that the parameter settings are better suited to each stage of the optimisation process than any static assignment, and likely to perform better than a deterministic schedule. The need for lengthy parameter experimentation is eliminated and practitioners do not have to have long-term experience in the field of optimisation. It seems that further research on many aspects of adaptive parameter control is well justified.

7.1. Continuous Parameter Values

Continuous domains have a potentially infinite number of values. The discretisation problem is largely a question of how small a change in a value can cause a meaningful change in algorithm performance. Little work has been done in this area, with the main contributions focussed on how to subdivide the feasible range into intervals. Adaptive Range Parameter Control (ARPC) [Aleti et al. 2012] adapts intervals during the optimisation run, and Entropy-based Adaptive Range Parameter Control (EARPC) [Aleti and Moser 2013] clusters parameter values based on their performance. More research is required to understand the effect of small changes in continuous parameter values. Ideally, the adaptation algorithm should accommodate differences in the desirable granularity of a parameter automatically.

7.2. Fitness Landscape Characterisation

Investigations into the properties of fitness landscapes have been motivated largely by the desire to identify the most suitable algorithm for a problem. Identifying the most suitable method still leaves the question of its parametrisation. The field of fitness landscape characterisation has yielded some scalar descriptors of landscape features [Moser et al. 2016; Aleti et al. 2016; Aleti and Moser 2015]. One example is the work by Reehuis et al. [2013, 2011], who established two metrics for estimating the *interestingness* and *novelty* of a solution. Interestingness is defined as the learning progress in the region that the solution lies in. Novelty depends on the similarity of a new candidate to previous solutions. The study finds that the best search strategy favours the most interesting solutions. Interesting solutions have sufficient—but not too extreme—novelty, which may be due to noise.

Measures of novelty and interestingness are frequently encountered in the area of developmental robotics, and have been widely researched in reinforcement learning [Schmidhuber 1991; Baranes and Oudeyer 2013; Cuccu et al. 2011]. They remotely relate to measures of diversity and convergence used in feedback collection strategies, since they are used for a similar goal: to ensure an appropriate balance between exploration and exploitation of the search space. A range of such metrics has arisen from fitness landscape analysis. Some appear suitable as feedback mechanisms for adapting parameter values.

7.3. Uncertainty and Parameter Interactions

Parameter-effect assessment methods associate specific performance properties of an EA with particular parameters, and vary the parameter settings according to the

changes in these properties. However, EAs are stochastic systems that may produce different result qualities for the same parameter values [DeJong 2007]. Moreover, the performance of the algorithm can be affected by more than one parameter value. Thus, using the performance improvement reported by the feedback strategy directly as an indication of the effect of specific parameter values may be misleading. The uncertainty regarding the actual cause of the improvement in the algorithm's performance and the stochastic nature of the EAs have to be accommodated by the parameter effect assessment strategy.

EAs have a number of parameters that influence the quality of the result. The quality achieved is likely to be the result of a combination of parameter values; each parameter can be assumed to contribute to a different degree at different stages. There is a need for a thorough investigation into the potential of integrating information about possible interactions between values of different parameters into the adaptive parameter-control strategy.

7.4. Application of Adaptive Parameter Control in Other Optimisation Methods

This survey focusses on adaptive parameter control for EAs, omitting studies that investigate the control of parameter values in other stochastic and approximate optimisation methods. Research in parameter control for other optimisation algorithms such as Local Search [Veerapen et al. 2012], Stochastic Local Search [Battiti et al. 2008], Ant Colony Optimisation [Lin and Zhang 2013; Stützle et al. 2012], Tabu Search [Wu and Hao 2013; Battiti et al. 2008; Battiti and Campigotto 2012] and Differential Evolution [Islam et al. 2012] demonstrate the efficiency of adjusting parameter values during the optimisation process in other approximate methods. Such investigations into the parametrisation of algorithms are most often conducted in isolation. An important future direction is the consolidation of the existing research efforts in order to extract common features and derive a common agenda for future research. It is likely that mechanisms applied in the parametrisation of one algorithm could be beneficial to the performance of another.

7.5. Evaluation of Adaptive Parameter Control Methods

Performing an experimental comparison between algorithms, and between EAs in particular, for the sake of evaluating parameter-control methods, poses a number of methodological questions, such as (i) How should the experiments be conducted? (ii) What benchmark problems and problem instances should be used? (iii) What benchmark parameter-control methods should be used? (iv) Which comparative measures should be reported? A unified test framework for adaptive parameter-control methods would greatly improve the comparability of different studies.

In the absence of such an experimental framework, identifying a diverse set of performance measures would improve the quality of experimental evaluations. If the optimal solutions of the problems considered are known, a Success Measure (SM) can be defined as finding the optimal solution(s). An SM can be used to define a performance measure as 'the percentage of runs terminating with success' [Eiben and Jelasity 2002].

Another conspicuous performance measure is the *Mean of Solutions Quality*, the quality of the final solutions achieved over a predefined number of function evaluations, or the *Average Number of Evaluations to a Solution* [Eiben and Jelasity 2002], the number of function evaluations used to achieve a certain quality. Reporting the variation in result quality across restarts would give an indication of the reliability of the approach. In addition to these obvious measures of quality, there is a need to quantify the overhead incurred by parametrisation method. Most studies use only a single performance measure in their comparisons. A more comprehensive set would greatly improve the information value of investigations into parameter control. Where

performances of approaches are compared, results should be validated using statistical tests, such as the Kolmogorov-Smirnov (KS) nonparametric test [Pettitt and Stephens 1977].

8. CONCLUSION

The number of studies into adaptive parameter control of EAs has approximately doubled since the year 2000. While the interest in mutation, crossover, and population has increased in line with this trend, the parent selection and replacement procedures, as well as solution representation, have received little attention.

The feedback collected to guide the parametrisation has been dominated by the phenotype of the solutions, assessed predominantly as the best current fitness or the fitness improvement compared to an immediate ancestor. A great number of approaches use a learning mechanism to attribute the quality achieved to a parameter value. Few approaches project the quality using a prediction model. Parameter values are mostly chosen in a greedy or quality-proportionate fashion.

Many open research questions of interest remain. Each of the four components of the conceptual model for parameter-control strategies has a number of variations that were implemented in different studies. A systematic comparison of their combinations would help identify which feedback mechanisms, effect assessment approaches, attribution methods and value update strategies are most effective in combination.

There is little doubt that the parameters interact in the optimisation. More sophisticated parameter-control methods may be able to incorporate the entropy between the parameters. Another challenge is the development of a more unified testing environment that employs a comprehensive set of measures to facilitate the comparison between the outcomes of different studies.

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